Highlights

WaveSplat: Grid-Free Wave Field Estimation from Sparse 3D Point Clouds
with a Learned Spectral Prior

Saeyong Park

- Grid-free wave estimation: continuous-wavenumber atoms with a learned prior.
- Beats the strongest eligible grid baseline on crowded synthetic seas.
- Survives a non-linear reference field and seeds outside the training corpus.
- Ablations attribute the gain to the prior and to wavenumber continuity.
- On measured records it does not reach the pre-registered decision threshold.

WaveSplat: Grid-Free Wave Field Estimation from Sparse 3D Point Clouds with a Learned Spectral Prior

Saeyong Park[a,*]

[a] *Avikus (HD Hyundai), Republic of Korea*

Abstract

Phase-resolved ocean wave fields must be reconstructed from sparse, irregular point clouds. The prevailing approach assimilates such returns into a spatial grid state, which requires interpolating them onto that grid; the closest recent study measures direct assimilation of the sparse returns instead and reports limited improvement at their original locations, while its outlook names a spectral-state formulation. We give that formulation: the field is a small set of continuous-wavenumber spectral atoms constrained by the dispersion relation, placed by a learned set-valued prior and refined by variable projection before per-frame updating. On crowded multi-component synthetic seas with a non-linear reference field and scene seeds drawn from outside the training corpus, it lowers median normalised RMSE by -40.2% against the strongest eligible grid baseline we possess (43/48 episodes), in a comparison where that baseline itself retains skill against a zero predictor (47/48). The claim is carried by ablation rather than by baseline comparison alone: removing the learned prior (45/48 episodes worse, against an analytic package that itself clears a zero predictor on only 15/48) and removing only the continuity of the wavenumber (46/48, dose-responsive) each cost measurable accuracy. The gains are synthetic and crowded-stratum only, the checkpoint was trained on the same distribution family so "out of sample" means on the seed axis alone, and on 12 measured fixed-station records the

*Corresponding author
Email address: saeyong.park@avikus.ai (Saeyong Park)

prior does not reach the pre-registered decision threshold against a per-record zero-field reference.
Keywords: wave field estimation, LiDAR, spectral methods, dispersion relation, data assimilation, sequential estimation

Table 1: Nomenclature.

Symbol	Description
$z(x, y, t)$	Surface elevation [m]
N	Number of spectral components
w_i	Amplitude of i-th component [m]
$k_{x,i}, k_{y,i}$	Wavenumber components [rad/m]
ω_i	Angular frequency [rad/s]
φ_i	Phase offset [rad]
g	Gravitational acceleration [m/s^2]
d	Water depth [m]
H_s	Significant wave height [m]
T_p	Peak period [s]
c_j	Confidence weight for observation j
θ_j	Beam incidence angle [rad]
r_j	Slant range to observation j [m]
M	Number of LiDAR observations per frame
nRMSE	RMSE normalized by H_s

1. Introduction

1.1. Motivation: wave field information for maritime autonomy

Table 1 summarizes the notation used throughout. Knowledge of the surrounding ocean wave field is a prerequisite for many functions in autonomous and semi-autonomous ship operations [1, 2]: short-term prediction for proactive

motion compensation during cargo handling and helicopter landing [3], route optimization to minimize fuel consumption and fatigue loading [4], wave-preview feedforward ride control on high-speed hydrofoil vessels [5], and structural health monitoring [6]. In each application the value of the estimate depends on its spatial resolution, update rate, and accuracy.

1.2. LiDAR point clouds for ocean surface sensing

Among sensor modalities, wave buoys [7] lack spatial resolution, X-band marine radar [8, 9] images large areas but measures backscatter rather than elevation, and satellite altimetry [10] is too coarse for vessel-scale real-time use. Scanning LiDAR, by contrast, provides *direct* elevation measurements over a wide area at $\sim$10 Hz [11]. That capability is not hypothetical: airborne scanning lidar has resolved open-ocean wave fields and their directional wavenumber spectra since the 1990s [12], from aircraft rather than from ships and at a correspondingly larger footprint. We note this explicitly because the absence of *shipboard lidar* validation in the present work (Section 6.6) reflects the absence of a *publicly archived* phase-resolved lidar sea-surface dataset, not the absence of such measurements. Current automotive-grade solid-state LiDAR systems operating at eye-safe wavelengths (1550 nm) offer $>$100 channels, wide fields of view ($\sim 120^\circ \times 25^\circ$), and ranges up to 250 m with centimeter-level accuracy in the near field. However, LiDAR observations are sparse (50–200 valid returns per scan), unevenly distributed (denser near the ship), and noisy (degrading with range and incidence angle). The challenge is therefore to reconstruct a continuous, physically consistent wave field from these sparse, heterogeneous observations in real time.

1.3. Grid-based paradigm and its limitations

The dominant approach follows a grid-based data assimilation paradigm adapted from numerical weather prediction: discretize space into a regular grid, represent the wave state at grid points, and apply an ensemble Kalman filter

(EnKF/ETKF) with FFT-based spectral forecasting [13, 3, 14]. While principled, this approach has three limitations under sparse LiDAR: (i) *interpolation artifacts*—sparse returns must be mapped to grid points, introducing smoothing near the Nyquist limit (5 m grid $\rightarrow \sim$10 m minimum wavelength regardless of LiDAR density); (ii) *fixed resolution* independent of scene complexity, wasting capacity on simple seas and under-resolving complex ones; and (iii) *forecast cost* $O(N_e G^2 \log G^2)$ per step regardless of true wave complexity, e.g. 50 independent 2D FFTs for N_e=50, G=40×30.

1.4. Proposed approach

We propose **WaveSplat**[1], a grid-free spectral representation that eliminates the spatial grid entirely. Instead of discretizing space, we represent the wave field as a weighted superposition of N plane waves:

$$z(x, y, t) = \sum_{i=1}^{N} w_i \cos(k_{x,i} x + k_{y,i} y - \omega_i t + \varphi_i) \tag{1}$$

where each spectral component is parameterized by amplitude w_i, wavenumber $(k_{x,i}, k_{y,i})$, and phase φ_i. The angular frequency ω_i is not a free parameter but is *constrained* by the linear dispersion relation:

$$\omega_i = \sqrt{g\,|k_i|\,\tanh(|k_i|\,d)} \tag{2}$$

where $g = 9.81$ m/s^2 and d is the water depth. This hard constraint reduces the parameter space from $5N$ to $4N$, regularizes optimization, and ensures physically plausible wave propagation during temporal tracking.

Our key contributions are:

1. **Grid-free spectral estimation framework (C1)**: Direct estimation on sparse, irregularly distributed observations without grid interpolation, enabling wave field evaluation at arbitrary spatial resolution. Unlike

[1]The name reflects "wave" + "splat" in the sense of projecting (splatting) spectral components onto arbitrary spatial locations, analogous to point-based rendering; it does not imply a methodological connection to 3D Gaussian Splatting [15].

single-snapshot sparse spectral methods [16], the spectral component representation persists across frames, allowing computationally efficient per-frame updates via dispersion-based phase prediction ($\sim$10 gradient steps) rather than re-running initialisation each frame, while supporting continuous monitoring without the discrete update latency of filter-based approaches. Estimation accuracy depends on initialization quality; tracking refines amplitudes and phases given the wavenumbers established during initialization.

2. **Dispersion-constrained parameterization (C2)**: Hard enforcement of the full dispersion relation (including tanh depth dependence) reduces the free parameter count and acts as a physics-based regularizer, improving tracking stability over extended time horizons. (Appendix Appendix A, R4.) The claim made here is the parameter-count reduction. Whether this constraint contributes to *accuracy* **has not yet been measured**: the ablation arm previously used to support it removed the frequency initialisation together with the constraint, and a controlled intervention that isolates the constraint was null in both strata.
3. **Geometry-aware confidence weighting (C3) — not claimed** (Appendix Appendix A, R1): Per-observation weights $c_j = \cos\theta_j/(r_j/d_{\text{ref}})$ based on incidence angle and range were introduced to account for the heteroscedastic noise structure of LiDAR measurements on ocean surfaces. This is **withdrawn as a contribution**: on a fresh out-of-sample corpus, disabling the weighting was *better* in both the sparse and the crowded stratum. It is retained below only as a reported negative result.

The pipeline also includes an online model-order mechanism (energy-based pruning and residual-triggered growth) that matches the number of spectral components to scene complexity, from N=2 for simple swells to N=21 for complex multi-component seas. We deliberately do *not* list it as a contribution: measured under the corrected tracker with a median-of-3 seed protocol, its accuracy benefit is null (retracted corpus; archived), and we discuss it as a mech-

Table 2: Comparison of ocean surface sensing modalities for vessel-scale wave field estimation. Typical values for operational systems.

Sensor	Spatial range	Resolution	Update rate	Direct elevation?	Key limitation
Wave buoy	Point	—	1 Hz	Yes	No spatial information
X-band radar	0.5–4 km	∼7.5 m	1–2 s/rev	No (intensity)	Requires SNR inversion
Satellite (altimeter)	Global	∼7 km	Days	Yes (nadir)	Coarse, non-real-time
Shipboard LiDAR	10–250 m	∼0.1 m (near)	10 Hz	Yes	Sparse, range-dependent noise
Ship motion (RAO)	—	—	Continuous	No (indirect)	Limited to encountered waves

anism and a compute-efficiency device in Section 6.6 instead.

Relative to prediction-error frameworks for deterministic wave forecasting such as LEPrE [17], which quantify the error of a linear-fitting prediction procedure without proposing an estimator, we contribute the estimator itself: a nonlinear, dispersion-constrained, grid-free fit with online model order, evaluated on scanning-LiDAR sampling geometry.

The remainder of this paper is organized as follows. Section 2 reviews related work on wave field estimation. Section 3 presents the WaveSplat algorithm in detail. Section 4 describes the simulation environment and evaluation metrics. Section 5 presents experimental results including baseline comparisons, ablation studies, and robustness analysis. Section 6 discusses the findings and limitations. Section 7 concludes.

2. Related Work

We organize the review around two axes: (1) the sensor modality used for ocean surface observation, and (2) the estimation algorithm used to reconstruct the wave field from observations. Table 2 provides a quantitative comparison of sensor modalities.

2.1. Conventional shipboard wave sensing

X-band radar is the most extensively studied sensor for shipboard wave estimation. Young et al. [9] first demonstrated that radar backscatter modulation (tilt, hydrodynamic, shadowing) carries wave spectral information, and

Nieto-Borge et al. [8] recovered the directional wave spectrum via 3D FFT in the (k_x, k_y, ω) domain with the linear dispersion relation serving as a spectral filter. This "dispersion filtering" concept is conceptually related to our dispersion-constrained parameterization, though applied in the Fourier domain rather than as a hard parameter constraint. Radar's fundamental limitation is that it measures backscatter intensity rather than surface elevation directly, requiring an empirical modulation transfer function that depends on radar parameters, wind speed, and sea state [8]; its spatial resolution ($\sim$7.5 m) is also coarser than LiDAR ($\sim$0.1 m in the near field). **Ship-motion-based** methods instead invert the vessel RAO to recover the incident wave spectrum from heave/pitch/roll measurements [18, 19]; these require no additional sensors but cannot resolve the spatial wave field, only the directional spectrum at the vessel location.

2.2. Grid-based wave field reconstruction

Deterministic wave field reconstruction from spatially resolved LiDAR/radar has been dominated by spatial discretization combined with ensemble data assimilation. Desmars et al. [13] introduced ETKF-based reconstruction from X-band radar using spectral propagation on a regular grid; the Ensemble Transform Kalman Filter [20] is preferred over the stochastic EnKF [14] for its deterministic analysis step. Desmars et al. [3] extended this framework with a nonlinear reconstruction-and-prediction scheme on remotely measured surfaces, demonstrating short-term forecasting up to one peak period ahead. Adaptation of ensemble Kalman filtering to scanning shipboard LiDAR has, to our knowledge, not been reported in the open literature; we therefore implement an ETKF-grid baseline by porting the radar-based formulation of Desmars et al. [13] to LiDAR observations (retracted corpus; archived). These grid-based methods share a common architecture—discretize the 2D domain into a regular grid, maintain an ensemble of spectral states, and assimilate observations via the Kalman update—which introduces an inherent trade-off between spatial resolution and computational cost, with interpolation from sparse observations to grid points being a known source of error. We distinguish this *determinis-*

tic reconstruction from the *stochastic* wave prediction performed by large-scale spectral models such as WAVEWATCH III [21], which forecast bulk statistics at kilometer scales without resolving individual phases.

A complementary line of work asks not how to reconstruct the field but how large the resulting prediction error is. Fucile et al. [17], in this journal, model the surface elevation as a Gaussian stochastic process and derive a semi-analytical Linear Estimator of Prediction Error (LEPrE), which returns the prediction-error standard deviation in time and space for a phase-resolved linear wave model whose coefficients are obtained by linear fitting on measurements, and which places the notion of a "Predictability Region" on a probabilistic footing; the formulation is verified against Monte Carlo simulation for long- and short-crested seas. Our contribution is orthogonal on three axes. First, in *what is produced*: Fucile et al. [17] characterise the error of a given deterministic prediction procedure and do not propose a field estimator, whereas we propose one and benchmark its reconstruction accuracy against ensemble-Kalman and Gaussian-process baselines. Second, in the *estimator class*: LEPrE presumes model coefficients determined by linear fitting, while WaveSplat obtains $(w_i, k_{x,i}, k_{y,i}, \varphi_i)$ by nonlinear joint optimisation with the dispersion relation imposed as a hard constraint and with the model order adapted online—a class outside the linear-fitting assumption on which the LEPrE derivation rests. Third, in the *evidence base*: the LEPrE formulation is agnostic to the measurement device, whereas our evaluation is specific to the sampling geometry of a scanning shipboard LiDAR, and that geometry is *simulated*: the shipboard configuration in this paper exists only in the sensor model. The measured records used in Section 5.6 come from fixed stations and from a different sensor modality, so no result here is supported by measured shipboard data, including sparse, range- and incidence-dependent returns and reflectivity-driven dropout. The two are naturally combined rather than opposed: WaveSplat currently returns point estimates without an error model (Section 6.6), and the LEPrE framework is the natural route to supplying one for the grid-free spectral representation.

2.3. Position of this work relative to the closest grid-based study

The closest study to ours assimilates sparse three-dimensional returns into a *spatial grid* state with an ensemble transform Kalman filter [22]. Two things about it shape our position.

First, sequential and continuously-updating operation is *not* what separates this paper from that line: recursive per-frame filters already exist there, and we therefore make no novelty claim on that axis. What differs is the state itself — a small set of continuous-wavenumber atoms rather than a grid — and the fact that the atoms are placed by a learned prior.

Second, and more usefully, that study does not merely leave direct sparse assimilation as future work: it *measures* it, and reports limited improvement at the original sparse locations relative to interpolating the returns first. Our contribution is therefore better described as a spectral-state implementation of an intervention whose grid-state version has already been tried and found wanting, than as an untried idea. We are careful not to over-read the outlook of that paper: it points at a spectral-state formulation, not at a learned prior or at continuous wavenumbers, and the ablations in Section 5.4 are what carry those two components here.

2.4. Grid-free alternatives: spectral, GPR, and learning-based

Several alternatives to grid-based assimilation have been explored. Classical **spectral estimation** methods (FFT, Welch, MUSIC, ESPRIT) assume regular sampling [23]; Capon's minimum-variance beamforming [24] and MUSIC [25] handle irregular arrays but either provide only power spectra without phases (Capon) or require known model order (MUSIC). WaveSplat leverages beamforming for *initialization* and refines the estimates by gradient optimization with physics constraints, combining beamforming robustness with joint amplitude-phase recovery. **Gaussian process regression** [26] with Matérn or squared-exponential kernels provides nonparametric spatial interpolation but scales as $O(M^3)$ and, in its standard single-frame form, lacks a temporal propagation model; sparse approximations [27] mitigate cost but still treat the wave

field as a generic spatial random process rather than a dispersion-constrained propagating signal. **Learning-based** methods include physics-informed neural networks [28, 29], the U-Net/Fourier neural operator of Ehlers et al. [30] for phase-resolved reconstruction, and the nonlinear ETKF extension of Desmars et al. [3]. These approaches show strong results on their training distributions but face practical challenges for arbitrary shipboard sensor configurations: FNO-style models assume uniform training grids incompatible with scanning-LiDAR irregular sampling, generalization across sensor geometries is limited, and the black-box nature complicates failure diagnosis. WaveSplat, by contrast, handles irregular observation geometries natively and produces directly interpretable spectral parameters that satisfy the dispersion relation by construction. It does use a learned component — the set-valued prior that proposes the initial atoms — but the learning enters only as a *proposal* over a physically constrained parameter space, not as a grid-shaped regression target: the estimate that leaves the pipeline is a set of dispersion-consistent atoms fitted to the observations, and the ablation in Section 5.4 is what measures how much the prior contributes. Direct experimental comparison with learning-based methods is beyond the scope of this work, which focuses on physics-constrained spectral estimation; such a comparison would require careful control of training distribution, sensor geometry, and computational budget. Its grid-free sparse representation is conceptually related to the compressed sensing literature [31, 32] and to Donelan et al. [16]'s directional spectral analysis of propagating waves, with three key differences: (1) each basis function is a physically meaningful plane wave with frequency constrained by the dispersion relation rather than a generic sinusoid or wavelet; (2) the model order is adapted *online* during tracking based on residual energy, rather than fixed a priori; and (3) the spectral components *persist* across frames: because each atom carries its own ω, the forward model evaluates at any later time without re-estimation, so a frame costs 10 gradient steps rather than a fresh reconstruction. We state this as a structural property and not as a demonstrated latency, and we measure it rather than assert it (Section 3.5).

3. Method

3.1. Forward model

The wave field is modeled as a superposition of N linear plane waves (1). Each component i has four free parameters: amplitude w_i, wavenumber components $(k_{x,i}, k_{y,i})$, and phase offset φ_i. The angular frequency ω_i is deterministically computed from the dispersion relation (2), reducing the total parameter count from $5N$ to $4N$.

This linear forward model is an $O(\epsilon^2)$ approximation to higher-order wave theories (e.g., Gerstner waves or second-order Stokes), where $\epsilon = |k| \cdot |w|$ is the wave steepness. At Beaufort 4, typical steepness is $\epsilon \approx 0.05$, giving a model mismatch of $< 0.25\%$. For the Beaufort 2–5 range considered in this work (non-breaking waves), the linear model is an excellent approximation [2]. This is an analytic estimate for real ocean waves. It does **not** describe the synthetic corpus used in this paper: there, the ground truth is a displacement-free linear superposition, so the measured mismatch against the model equation is exactly zero (see Section 4).

3.2. Observation model

At each LiDAR scan at time t, we observe M elevation measurements $\{(x_j, y_j, z_j)\}_{j=1}^{M}$ at irregular spatial locations. The observation model is:

$$z_j = \sum_{i=1}^{N} w_i \cos(k_{x,i} x_j + k_{y,i} y_j - \omega_i t + \varphi_i) + \varepsilon_j \tag{3}$$

where ε_j is observation noise with variance depending on range and incidence angle. The number of valid observations per frame varies between 50 and 200, depending on sea state, vessel motion, and LiDAR raydrop characteristics.

3.3. Geometry-aware confidence weighting (C3), which we do not claim

This mechanism is described here for completeness, but it is *withdrawn as a contribution*: on a fresh out-of-sample corpus, turning the weighting off was better in both the sparse and the crowded stratum. Read this section as a

description of an implemented component and a reported negative result, not as a claimed accuracy benefit.

LiDAR elevation accuracy on ocean surfaces degrades with two physical mechanisms: (1) beam divergence increases the footprint area at longer ranges, averaging over local surface curvature; and (2) at high incidence angles (near-grazing geometry), the probability of specular reflection increases, causing either total signal loss or biased returns from wave crests.

We assign per-observation confidence weights:

$$c_j = \frac{\cos(\theta_j)}{r_j / d_{\text{ref}}} \tag{4}$$

where θ_j is the beam incidence angle between the LiDAR ray direction and the local water surface normal $\mathbf{n}$, r_j is the slant range to observation j, and $d_{\text{ref}} = 125$ m is a reference distance chosen as half the maximum LiDAR range. In the simulation experiments reported here, $\mathbf{n}$ is obtained from the ground-truth surface mesh, which (see Section 4) is a displacement-free linear superposition rather than a Gerstner surface. For real-data deployment, the surface normal is unknown prior to reconstruction; a pragmatic approximation is $\mathbf{n} \approx \hat{\mathbf{z}}$ (flat-surface assumption), introducing a slope-dependent error bounded by the wave steepness $\epsilon \approx 0.05$ at BF4 ($\sim 3^\circ$ normal deviation). For near-normal observations ($\theta < 30^\circ$) this yields $< 2\%$ error in $\cos\theta_j$; grazing observations ($\theta \approx 60^\circ$) can see up to $\sim 9\%$ error. These weights (4) are incorporated in the optimization loss as weighted mean squared error, giving higher influence to close-range, near-normal observations.

3.4. Two-phase estimation

The estimation proceeds in two phases: an initialization phase that establishes the spectral model from multi-frame observations, followed by a per-frame tracking phase that maintains the model with minimal computation. Table 3 and Fig. 1 summarize the complete pipeline.

Table 3: Algorithm: the deployed estimation pipeline, transcribed from the shipped configuration (`init_method`=`auto`, `auto_switch_thresh`=0). Constants are the values in the source, not targets. *This replaces an earlier listing that described the analytic initialiser*: under the shipped configuration the router resolves to the learned path, and the analytic beamforming initialiser and its N=3 → 8 → 15 staging are not reached. The analytic procedure is still described in Section 3.4.1 because it is the comparison arm of the prior ablation.

Phase 1: Initialization (frames with $t \leq 2T_p$; see the note on T_p in Section 3.4.1)
1. Accumulate the window's observations $\{(x_j, y_j, z_j, t_j)\}$
2. Subsample to at most 400 points, fixed seed — *for the proposal only*
3. Conditioning map: 64×64 beamform power over $k_x, k_y \in [-0.6, 0.6]$ rad/m, ω from the dispersion relation
4. Conditional diffusion: 4 draws × 32 atoms, 500 reverse steps
5. Per-draw polish: 80 Adam steps on the phase-marginalised likelihood
6. Pool: keep energy $> 0.03\, e_{\max}$; *drop* any atom within Δk=0.03 rad/m of a stronger one; cap at $N_{\max}$=30
7. Amplitude and phase in closed form: ridge least squares (λ=0.5)
8. Joint non-linear fit: *disabled* (0 iterations)
9. Adaptive order: prune to 90% energy (roll back if residual RMS rises >30%); if residual RMS > 0.05 m, add via a 61×61 scan with 31×31 refinement, random initial phase, then 50 Adam steps
10. Variable projection: 400 Adam steps on (k_x, k_y) with the linear coefficients eliminated in closed form; steepness clamp 0.15
Phase 2: Tracking (each subsequent frame)
1. Slide a K_w=3 frame window
2. Freeze $(k_{x,i}, k_{y,i})$; 10 Adam steps at lr 0.005 on (w_i, φ_i) over the window — loss: confidence-weighted MSE + $\lambda_1 \|w\|_1$ (λ_1=10^{-4}) + spectral smoothness
3. Steepness clamp 0.15 — *no explicit phase advance*: the forward model's $-\omega_i t$ term carries the propagation

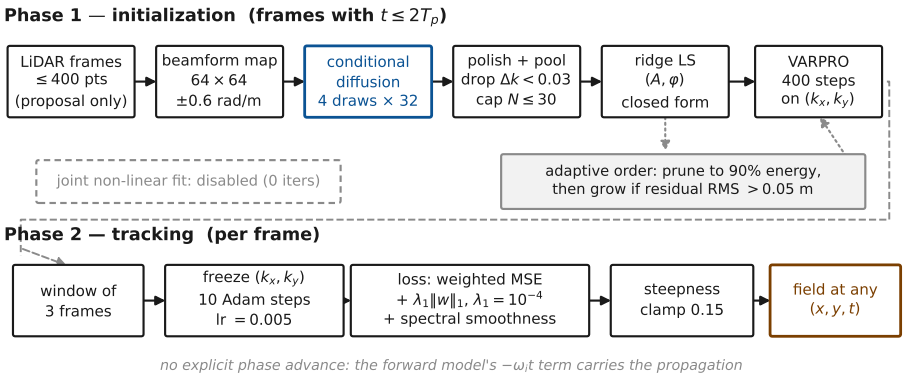


Figure 1: The deployed pipeline. Phase 1 conditions a diffusion prior on a coarse beamform map of the initialisation window, draws and polishes candidate atom sets, pools them into at most 30 atoms, fits amplitude and phase in closed form, and refines the wavenumbers by variable projection; the adaptive prune-and-grow step runs before variable projection, on a model that has had no joint refinement. Phase 2 freezes the wavenumbers and updates (w_i, φ_i) over a three-frame window. Blue marks the one learned component, brown the output. The dashed rectangle is a stage that exists in the code and is switched off in the shipped configuration; the dashed path is the transition between phases. Every constant printed in this figure is read from the source at generation time by `scripts/gen_fig_pipeline.py`, which also asserts the four switches that would change its structure.

3.4.1. Phase 1: Initialization

Given the first $T_{\text{init}} = 2T_p$ seconds of observations, we must place N atoms in the (k_x, k_y) plane and fit their amplitude and phase. Two procedures do this in our system, and *which one runs is the subject of one of our ablations*, so we describe both.

The learned prior: architecture, training data, and what we cannot certify. The prior is a conditional denoising diffusion model over *sets* of atoms. Its conditioning branch is a small convolutional encoder that reduces the 64×64 beamform map to 256 tokens of width 128 (3×3 convolutions, group normalisation, GELU, with a learned positional embedding). Its denoiser is a stack of 4 pre-norm transformer blocks of the same width: within each block, self-attention over the 32 atom slots, cross-attention into the conditioning tokens, and a two-layer

feed-forward network. Atoms enter and leave as three normalised numbers — two wavenumber components and a log-energy — through linear projections, so the model is permutation-equivariant over the set by construction and has no notion of atom ordering. It has 1,454,723 parameters. Sampling is ancestral DDPM with a cosine noise schedule over 500 steps, and training is the standard ϵ-prediction mean-squared error under AdamW (learning rate 2×10^{-4}, weight decay 10^{-4}, cosine annealing, batch 32).

Training scenes are synthetic and are drawn from a deliberately stratified mixture rather than from the evaluation distribution: discrete crowded (15 and 30 components) at 0.30, discrete sparse (1 and 2) at 0.20, five-component at 0.10, two continuum strata at 0.10 and 0.15, and an intermediate-spreading stratum at 0.15, with sea states covering Beaufort 2–5 anchors and a jittered box around them. The observation geometry used in training replays frame-structured sweep windows rather than drawing observation times independently. *Scene seeds for training are drawn from a band disjoint from every seed used by any evaluation generator in this paper*, so no evaluation episode was seen during training, although the generator family is the same — which is the sense in which Section 5.7 says only the seed axis is out of sample. The shipped checkpoint was trained for 40,000 steps from scratch, and the estimator refuses to load a checkpoint whose SHA-256 does not begin `636c72a797ee`.

Two things about this we cannot certify and will not claim. First, the six training strata were measured only as a bundle: no ablation isolates what each contributes, so we cannot say which part of the mixture the prior's behaviour comes from. Second, the batch that produced this checkpoint recorded checksums for its own training script and preregistration, and those checksums do not match the files that were committed with them. The files themselves have not drifted — they are byte-identical to their single committed version — so what is missing is not the code but the guarantee: the freeze was taken, the files were then edited, and the freeze was never re-taken. The architecture and the checkpoint identity above are read directly from the shipped weights and are therefore certain; the training recipe is transcribed from that batch's report

and is *not* independently verifiable from the repository as it stands. A reader reproducing this work should treat the training procedure as documentation rather than as a frozen artefact.

A dependency we disclose rather than describe away. The length of this window is set by the true peak period of the episode, read from the scene metadata, rather than estimated from the observations; on the measured records it is read from the record's own reported spectrum. This is a property of the harness and not of any arm: the window is computed once, before an initialisation procedure is selected, so every arm in every paired comparison reported here sees exactly the same frames, and the scoring frames are chosen the same way. It does mean that a deployment would need to supply T_p from another source, and that no experiment here measures the sensitivity of the method to getting it wrong.

What "the shipped configuration" selects, and when. The router that selects an initialisation procedure is not purely a quality heuristic: it also branches on whether the input carries a ground-truth field. When none is present — which is the case for every measured record — the shipped configuration switches to a separate estimator built around a bank of candidate spectra, and the learned path described above is not entered at all. Table 3 therefore describes what runs on the synthetic corpora. The measurements that report no skill on the measured records (Section 5.6 and Table 7) were obtained by *overriding* that router and forcing the learned path, because the question those batches ask is what the learned prior does on real data, not what the shipped default does. The bank estimator is outside the scope of this paper and none of its results are reported here; we state its existence because a reader reproducing "the shipped configuration" on a real record would otherwise get a different algorithm than the one we describe.

Deployed path: a learned set-valued prior. The configuration evaluated in Section 5 places the atoms with a learned prior over wavenumber–energy *sets*. A conditional diffusion model, trained on synthetic scenes, is conditioned on the

initialisation-window observations and draws several posterior samples of an atom set; the pooled draws are then deduplicated in wavenumber thresholded on energy, then deduplicated in wavenumber (an atom is dropped when a higher-energy atom lies within $\Delta k = 0.03$ rad/m of it *on both axes*), and truncated at $N_{\max}$. Given the resulting wavenumbers, the amplitude and phase of every atom are obtained in closed form by ridge-regularised least squares, and the whole set is refined by *variable projection*: because the model is linear in (a_j, b_j) once $(k_{x,j}, k_{y,j})$ are fixed, the linear coefficients can be eliminated analytically and only the wavenumbers are iterated.

The learned proposal cannot place an atom outside a bounded region: the network emits normalised coordinates that are clipped to $[-1, 1]$ and rescaled by a constant stored in the checkpoint, giving $|k_x|, |k_y| \leq 1.1$ rad/m. For the sea states evaluated here the largest true wavenumber is near 1.0 rad/m, so the bound is satisfied with little margin, and it is a property of the checkpoint rather than of the representation — the analytic alternative searches ± 4 rad/m. A tighter bound applies upstream: the only view of the data the network receives is the conditioning periodogram, and that grid spans ± 0.6 rad/m. Components with $|k|$ between 0.6 and the clip bound are therefore outside the network's input entirely, and can be reached only by the residual-growth step, whose beamformer does search the wider range. We have not measured how often that happens. Any application to shorter waves would need the prior retrained, and we have not measured what happens when a true component lies outside the bound.

Two deployment choices matter for reproducing the numbers in this paper. Variable projection is enabled (400 steps); and the joint non-linear fit that earlier versions ran after initialisation is *disabled* (0 iterations) — it was measured to be redundant once variable projection is present. These two are not independent: removing the joint fit leaves a larger residual, which in turn makes the adaptive growth rule of Section 3.4.2 fire far more often than it did before. A third value in the shipped configuration is worth naming because it is *inert*: the configuration sets an analytic-path ensemble of 3, but the router resolves to the learned path and the ensemble size is then re-read for the resolved route, which ships at 1.

No median-of-k ensemble therefore runs in any synthetic result reported here. We say so because a reader inspecting the configuration file would reasonably conclude otherwise. We note this because it is easy to reason about either switch alone and be wrong about the resulting system.

Alternative path: dispersion-aware beamforming. The system also contains a purely analytic initialiser, which is what runs when the learned prior is not used. It is described next both because it is the fallback and because it is the *comparison arm* of the prior ablation in Section 5.4: replacing the learned proposal with this procedure, changing nothing else, is what accounts for a margin of -36.3% — against an analytic package that itself clears a zero predictor on only 15/48 episodes.

Dispersion-aware beamforming. Initial wavenumber estimates are obtained by scanning a Cartesian grid in wavenumber space. The grid covers $k_x, k_y \in [-4.0, 4.0]$ rad/m with 81×81 bins (resolution $\Delta k \approx 0.10$ rad/m), spanning wavelengths from ∼1.6 m to ∼130 m and encompassing Beaufort 2–5 conditions at d=50 m depth. The search bound of 4.0 rad/m is set conservatively above the highest wavenumber component present in the dataset ($|k|_{\max} \approx 1.0$ rad/m for BF3–5 sea states), ensuring full coverage with margin.

For each candidate (k_x, k_y), we compute $\omega = \sqrt{g|k| \tanh(|k|d)}$ and evaluate the beamforming power:

$$P(k_x, k_y) = \left| \sum_j z_j \, e^{-i(k_x x_j + k_y y_j - \omega t_j)} \right|^2 \tag{5}$$

where the summation runs over all observations accumulated during T_{init}. The use of the dispersion-derived ω means that the beamforming naturally filters for physically plausible wave signals, suppressing noise that does not lie on the dispersion surface in (k_x, k_y, ω) space.

Peaks in P are extracted greedily: the maximum is selected, then all candidates within a Cartesian radius of 0.15 rad/m are suppressed, and the process repeats. The 0.15 rad/m threshold prevents double-counting of the same wave component spread across adjacent wavenumber bins, while being narrow

enough to resolve crossing sea systems. The initial amplitude and phase of each extracted component are estimated via linear least-squares projection of the accumulated observations onto the selected wavenumber basis functions.

Each extracted peak is then refined locally via a second beamforming pass on a finer 41×41 grid spanning ± 0.2 rad/m around the coarse peak ($\Delta k_{\text{fine}} = 0.01$ rad/m), improving wavenumber accuracy from ~ 0.10 to ~ 0.01 rad/m. Without this refinement, the 0.10 rad/m coarse grid would limit $|\mathbf{k}|$ accuracy to roughly 17% at typical BF4 wavenumbers ($|\mathbf{k}| \approx 0.6$ rad/m), inducing significant phase drift during subsequent tracking.

Coarse-to-fine optimization. Starting from the beamforming initialization, we optimize the spectral parameters in three stages (N=3 $\rightarrow$ 8 $\rightarrow$ 15), with up to 200/200/500 iterations of the Adam optimizer [33] (learning rate $\text{lr} = 1 \times 10^{-2}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) at each stage. The loss function is:

$$\mathcal{L} = \underbrace{\frac{1}{M} \sum_j c_j \left(z_j - \hat{z}_j\right)^2}_{\text{weighted MSE}} + \underbrace{\lambda_1 \sum_i |w_i|}_{\text{sparsity}} + \underbrace{\lambda_s \sum_i (w_{\sigma(i)} - w_{\sigma(i-1)})^2}_{\text{spectral smoothness}} \tag{6}$$

where $\lambda_1 = 10^{-4}$ promotes amplitude sparsity (driving negligible components toward zero for subsequent pruning) and $\lambda_s = 10^{-4}$ penalizes abrupt amplitude variations between spectrally adjacent components (σ denotes ordering by $|k|$), encouraging smooth spectral envelopes consistent with JONSWAP-like spectra.

During optimization, wavenumber components $(k_{x,i}, k_{y,i})$ are free parameters (unlike in the tracking phase). The dispersion constraint is enforced by computing ω_i from $(k_{x,i}, k_{y,i})$ at each forward evaluation rather than treating ω_i as a free parameter, ensuring that the constraint is satisfied exactly throughout optimization.

3.4.2. Adaptive model order

After initialization, we apply energy-based pruning. Components are sorted by energy $E_i = w_i^2$ in descending order, and the minimum set capturing 90% of total energy $\sum_i E_i$ is retained. This criterion is motivated by the JONSWAP spectral shape, where a small number of components near the spectral peak

typically contain the majority of the wave energy.

A safety check prevents aggressive pruning from discarding physically significant components: if the post-pruning RMSE exceeds 130% of the pre-pruning value, pruning is rolled back entirely. This can occur when two components of similar energy contribute to different spatial regions (e.g., crossing seas), making each individually appear low-energy.

If the residual RMS after pruning exceeds $\tau_R = 0.05$ m (indicating unmodeled wave energy), new components are grown by applying beamforming to the residual signal $r_j = z_j - \hat{z}_j$ and adding the dominant residual peaks to the model, up to $N_{\max} = 30$. Grown components are initialized via this residual beamforming step and are not re-run through the full $N{=}3{\to}8{\to}15$ coarse-to-fine schedule, trading some initialization quality for computational efficiency at this stage. The threshold is derived from the theoretical residual floor: combining LiDAR range noise ($\sigma_r \approx 0.02$ m) with a nonlinear–linear model mismatch allowance at BF4 ($O(\epsilon^2)\, H_s/2 \approx 6$ mm; this term is an allowance for real waves and is **absent** in the synthetic corpus, where the measured mismatch is zero) and angular-noise projection ($\sim$5 mm) yields a lower-bound RMS of $\sim$0.022 m; $\tau_R = 0.05$ m provides a $\sim$2.3$\times$ safety margin for unambiguous detection of unmodeled signal. On the raycast-mesh-rendered dataset used throughout this paper, however, the mesh's own facet-rendering error contributes comparably to this budget (mean $\approx$0.048 m, measured directly for BF3–BF5 multi-component scenes on the shipped 40$\times$30 mesh, versus the $\sim$0.022 m physical floor derived above); part of what currently triggers residual-based growth may therefore be unmodeled rendering error rather than unmodeled wave content, and τ_R should be re-validated on a Nyquist-adequate mesh. We also note that τ_R is an *absolute* residual RMS in metres, compared without normalization by sea state, so its effective strictness varies with H_s; a scale-invariant form that normalizes the residual by the observed signal RMS is implemented but not enabled by default. A sensitivity sweep over $\tau_R \in \{0.02, 0.05, 0.08, 0.10, 0.15\}$ m on 12 representative episodes of the retracted corpus found no ordering among values in [0.05, 0.15] m produced indistinguishable average error, while $\tau_R = 0.02$ m caused over-growth

in multi-component seas. The per-value levels from that sweep are not quoted: it was run on the retracted corpus and before the tracker phase-advance correction, and no re-measured substitute exists. The value $\tau_R = 0.05$ m has been held fixed in every batch reported here.

3.4.3. Phase 2: Tracking

For each subsequent LiDAR frame we perform a single update step. There is no explicit predict step: an earlier version advanced the phases by $\varphi_i \mathrel{+}= \omega_i \Delta t$ before each update, but that path is switched off in the shipped configuration and the forward model's own $-\omega_i t$ term carries the propagation instead, with t the absolute frame time. The update performs $K = 10$ Adam steps (lr $= 0.005$) on the weighted MSE loss over a sliding window of $K_w = 3$ recent frames, updating only amplitudes w_i and phases φ_i while freezing wavenumbers $(k_{x,i}, k_{y,i})$ (`w_phi` tracking mode).

Freezing wavenumbers during tracking is motivated by two considerations. First, wavenumber estimation from a single frame is ill-conditioned: the LiDAR's irregular sampling does not constrain the wavenumber from one snapshot, whereas the initialisation window spans $2T_p$, which at 10 Hz and the peak periods used here is on the order of one to two hundred frames. Second, with the wavenumbers fixed the per-frame problem is linear in a small number of parameters, so a short optimiser budget suffices.

The sliding window ($K_w = 3$) uses the three most recent frames for gradient computation. This provides temporal averaging that smooths single-frame noise without introducing excessive latency: at 10 Hz, the window spans 0.3 s, much shorter than the peak period T_p for all sea states considered.

3.5. Computational complexity

Let M denote the number of LiDAR observations per frame and N the number of spectral components.

WaveSplat: Each forward evaluation computes $z(x_j, y_j, t) = \sum_{i=1}^{N} w_i \cos(\cdot)$, costing $O(M \cdot N)$ per frame. With K gradient steps and window size K_w, the

per-frame tracking cost is $O(K \cdot K_w \cdot M \cdot N)$. In practice, K=10, K_w=3, $N \leq 30$, measured on CPU. An earlier absolute per-frame figure quoted here was withdrawn as a cross-process timing (Appendix Appendix A, R3); the measured replacement is below.

This analysis covers the tracking phase only. It is unchanged by which initialisation ran, because the tracking loop does not branch on the initialiser, and we verified that: across a pre-registered 64-run batch spanning two initialisation routes, two strata, and CPU-only versus GPU-available execution, the per-frame tracking cost varied only between 5.9 and 6.3 ms.

Measured budget. On a single thread of an AMD Ryzen 7 7800X3D, in one process, with CPU-only execution and on the crowded stratum — the most conservative cell, fixed in the preregistration before results existed — the deployed learned path costs 5.91 ms per tracking frame, or 5.9% of a 10 Hz frame budget. Initialisation is a one-off 5.18 s, which is roughly fifty frame intervals: real for a cold start, and not part of the per-frame figure. Two measurements from that batch run against the intuition a reader may bring. First, the learned initialiser is *faster* than the analytic alternative, not slower —5.18 s against 26.61 s on the same episodes — despite running a 500-step reverse diffusion sampler four times; we report the direction and do not attribute it. Second, a GPU buys almost nothing here: the same initialisation takes 4.86 s with one available. Neither number is comparable with any latency in the earlier version of this paper, which came from separate processes, and the batch did not run a baseline arm, so no comparative timing claim is available.

A second, query-side timing. Evaluating the estimated field at a point costs 137 μs, linear in the number of query points over the measured range, so a 0.1 s frame budget admits of order several hundred query points. Two caveats travel with that number and neither is small: it was measured on the grid-based path used for the measured records and has *not* been re-measured with the learned renderer, and it covers the steady state after initialisation only, so the proposal stage — four diffusion draws of 500 reverse steps each, plus polish — is not in it.

Table 4: Computational complexity comparison. M: observations/frame, N: spectral components, G: grid size, N_e: ensemble members. ETKF uses the tuned configuration ($N_e = 100$, $G = 40 \times 30$). The "Measured" column is empty by design: the figures that stood there were withdrawn as cross-process timings (Appendix Appendix A, R3), and the same-process replacement measured since covers our own arm only (Section 3.5), not the baselines. The asymptotic column is analytic and is unaffected.

Method	Per-frame complexity	Measured
WaveSplat	$O(K \cdot K_w \cdot M \cdot N)$	[retracted]
ETKF-grid	$O(N_e \cdot G^2 \log G^2)$	[retracted]
GPR	$O(M^3)$	[retracted]

It is a budget for querying a field one already has, not a latency for producing one.

ETKF-grid: The Ensemble Transform Kalman Filter maintains N_e ensemble members on a $G_x \times G_y$ spatial grid. Each update involves FFT-based propagation at $O(N_e \cdot G_x G_y \cdot \log(G_x G_y))$ plus the ensemble transform at $O(N_e^3 + N_e^2 \cdot M)$. The tuned configuration is (N_e=100, G=40 × 30); see Table 4. No absolute latency is quoted for this arm: the earlier figure was withdrawn as a cross-process timing (Appendix Appendix A, R3) and no baseline arm has been re-timed.

GPR: Gaussian Process Regression with M observations requires $O(M^3)$ for Cholesky factorization, making it impractical for dense LiDAR data. Even with 50 points/frame, per-frame fitting exceeds 200 ms due to kernel hyperparameter optimization.

Table 4 summarizes the scaling comparison.

4. Experimental Setup

4.1. Simulation environment

We develop a physics-based LiDAR wave simulation (WaveSim) generating ground-truth wave fields with JONSWAP spectrum sampling [34]. The simula-

tor parameters are:

- **Sea states**: Beaufort 2–5 (H_s=0.3–2.0 m, T_p=4–9 s)
- **Wave complexity**: 1, 2, 5, 15, 30 spectral components with directional spreading (±20°)
- **Ground-truth wave model** (Appendix Appendix A, R2): The synthetic ground truth is a *displacement-free linear superposition on a fixed grid*: the generator discards the horizontal displacement components of the Gerstner solution, so the ground truth lies in the *same function space* as the estimator's model equation. Nonlinear validation was carried out in a separate higher-order-spectral (M=3) batch, not on this corpus.
- **LiDAR model**: 500 horizontal beams per scan, 10 Hz scan rate, range 10–250 m. The beam count represents the total number of ray directions per frame (horizontal fan); real-world solid-state LiDARs such as the Seyond Falcon K1 (1550 nm, 152 vertical channels, 120° × 25° FOV) produce comparable or higher point densities per scan
- **Raydrop models**: two return-sampling patterns that differ in how return density is distributed over range and incidence angle. (a) *Geometry-weighted*: retention probability $P \propto R_{\mathrm{Fresnel}}(\theta)\cos\theta/d^2$, so near-range, near-normal facets are preferentially retained. (b) *Unweighted*: each ray hit is retained with equal probability, calibrated to the same expected return count. We retain the historical labels "uniform" and "slope" for these two models for consistency with the released dataset. The second model was originally specified as a specular-backscatter channel, but that channel is unreachable at the grazing incidence angles of a mast-mounted LiDAR: retro-specular return requires the ray to strike within $\alpha/2 = 2.5°$ of the surface normal, whereas the beam fan spans 78–90° from vertical and wave slopes reach only ∼7° at BF7. The implementation consequently always takes its unweighted fallback path, which is what we describe and

report here. The contrast is therefore a test of sensitivity to the *spatial weighting* of returns, not of diffuse versus specular optics

- **Water depth**: $d = 50$ m (intermediate to deep water for the wavelengths considered)

Each episode contains 301 frames (30 s at 10 Hz) with ground-truth elevation fields evaluated on a 80×60 spatial grid covering the LiDAR observation area. A total of 100 episodes are generated across 5 random seeds per configuration. Each batch reported in Section 5 draws its own episode set from this corpus or from a corpus regenerated the same way, and states its own n; the comparisons are paired per episode and read with a sign test, with the direction and the decision threshold fixed before the runs. The earlier design — a five-metric panel over all 100 episodes, a twelve-episode per-episode breakdown, and a 100-run Wilcoxon validation — belongs to the retracted corpus and is not used here.

4.2. Mesh adequacy of the reference field

The ground-truth elevation field is rendered by ray-casting against a triangulated surface, so the mesh spacing sets an upper bound on the wavelengths the reference field can represent, independently of the wave model that generated it. The originally released episodes used a 40×30 mesh over the observation area, giving a spacing of 6.15×6.90 m. Applying the standard sampling criterion (at least two, and preferably six, mesh points per wavelength) to the true spectral components of each episode, *none* of the released episodes is adequate at its peak wavelength, and the calmest episodes are the worst affected: a BF2 single-component episode with $\lambda = 8.2$ m is rendered at 1.2 points per wavelength, below the Nyquist rate. Short-wavelength energy is therefore attenuated or absent in the reference field, which both understates the wave content an estimator could in principle recover and inflates the apparent error of any method that recovers it.

We regenerated the full 100-episode dataset on a 200×167 mesh (1.21 m spacing, adequate for every component present), holding the random seed fixed so that the wave components of each regenerated episode are bit-identical to its released counterpart. The two datasets are therefore paired, differing only in rendering. The synthetic batches of Section 5 are computed on this regenerated corpus or on one produced the same way. Three groups of results are not: the non-linear replication, which uses a different reference field and seeds drawn from outside it; the sea-state experiment, which consumes fresh seeds; and every measured-record result, which uses no simulator at all. The effect is strongly regime-dependent: sparse episodes improve substantially (up to −77% nRMSE, with previously undetected short-wavelength components recovered), whereas crowded 15–30-component episodes move much less, indicating that the crowded-regime error is not a rendering artifact. We report this explicitly rather than silently re-running, because two conclusions in earlier versions of this work—a claimed low-SNR failure mode (retracted corpus; archived) and a claimed beamforming detection threshold (Section 6.6)—were attributions of a rendering limitation to physics and to the algorithm respectively.

Tracker phase-advance correction.. A second defect, in the estimator rather than the reference field, was found and fixed in the same review, and several tables in this paper carry a provenance note referring to it. The tracking loop advanced each component's phase by $\varphi_i \mathrel{+}= \omega_i \Delta t$ at the start of a frame, while the forward model of Eq. (1) already carries the $-\omega_i t$ term with absolute episode time. The advance was therefore applied twice, so the represented field drifted out of step with the truth at a rate proportional to ω_i even when every parameter was correct. Removing the explicit advance (the current default) lowers the full-dataset nRMSE, raises the Skill Score and the variance ratio, and reverses the sign of the supplementary directional R^2. We do not quote the levels: they come from a corpus in which the two arms were never scored on the same frames, and only the *within-arm* direction of this correction survives that defect. Because the defect scaled with $\omega_i \Delta t$ it penalised short components

hardest, i.e. in the same direction as the mesh inadequacy above—which is why several conclusions in earlier drafts had two independent reasons to be wrong. Tables produced before this correction are individually marked; where a re-run has since been performed we report the re-run and say so, and where it has not we mark the table rather than mix eras silently. We note explicitly that the correction, the mesh replacement, and a move from single-seed to median-of-3 reporting all landed within the same revision, so differences between a pre-correction and a post-correction table cannot be attributed to any one of the three.

4.3. Metrics

We report:

- **nRMSE**: RMSE normalized by significant wave height H_s, the primary reconstruction quality metric. Lower is better; values below 0.10 indicate excellent reconstruction, 0.10–0.30 is good, and above 0.30 indicates significant error.
- **Inference speed**: ms/frame for the tracking phase (defined here for completeness; the measured value for the deployed path is in Section 3.5, and *no timing in this paper is comparable with any baseline* because no baseline arm was timed), measured on a single CPU thread.
- $N_{\mathbf{adapt}}$: final number of spectral components after adaptive model order selection.

Choice of metric and complementary measures.. We use nRMSE as the primary metric in the main results for direct comparability with prior LiDAR wave estimation work [13, 3]. However, we acknowledge that nRMSE alone has a known limitation: a trivial constant-zero predictor on a zero-mean wave field with $\sigma_z \approx H_s/4$ achieves nRMSE ≈ 0.25, so methods reporting nRMSE near this level cannot be distinguished from a flat-output predictor by nRMSE alone. This is particularly relevant when comparing methods that may under-fit (e.g.,

over-smoothed Gaussian process regression) against methods that capture wave structure accurately.

To address this limitation, we define a complementary *pre-specified metric panel* of four secondary metrics that expose under-fitting and quantify physics-prior contribution: (i) variance ratio VR = $\mathrm{Var}(\hat{z})/\mathrm{Var}(z)$ (amplitude preservation; VR$\ll$1 indicates under-fitting); (ii) Pearson correlation ρ between $\hat{z}$ and z (shape match independent of amplitude); (iii) skill score SS = $1 - \mathrm{MSE}(\hat{z})/\mathrm{MSE}(z_{\mathrm{ref}})$ relative to a zero predictor reference (SS$\leq$ 0 means the method is no better than predicting zero); and (iv) $\mathrm{RMSE}_{\mathrm{unobs}}$, the RMSE evaluated only on grid points that lie at least 10 m from any LiDAR observation (quantifies how well the method reconstructs the wave field where no direct observation exists, isolating the contribution of the physics prior). All five metrics are defined in our public evaluation module. *This paper reports nRMSE only.* The four secondary metrics were part of the retracted corpus's evaluation panel and were not re-measured for the batches reported here, so we describe them for completeness and do not claim a five-metric evaluation. Each batch's primary metric was fixed in its preregistration before results existed, which is the protection against metric selection that we actually have.

4.4. Baselines

ETKF-grid [13]: Ensemble Transform Kalman Filter on a regular grid with FFT-based spectral propagation, multiplicative and additive inflation, and observation thinning to $M_{\max}$=40 per frame to keep the observation-state projected matrix numerically well-conditioned and bound per-frame computational cost. To avoid penalizing the baseline by an arbitrary parameter choice, we performed a 24-configuration hyperparameter sweep over grid resolution {40×30, 80×60}, ensemble size {50, 100}, and inflation pairs (α, σ) covering the standard ranges recommended by Desmars et al. [13]; the winning configuration (g=40×30, n_{ens}=100, α=1.0, σ=0.005) is used as "ETKF (tuned)" wherever an ETKF arm appears. The qualitative structure of the sweep was that the coarser 40×30 grid outperformed 80×60, the finer grid leaving the ensemble undersized

for the available observations, and that minimal multiplicative inflation won given the additive inflation already present.

The per-configuration error levels from that sweep are not quoted here: they were measured on the corpus whose cross-arm comparisons are retracted, and no re-measured substitute exists. No ETKF result appears in Section 5; the arm is retained in this section because the sweep is part of how the baselines were chosen, not because a number from it is used.

Spectral-background LETKF (the headline comparator): a localised ensemble transform Kalman filter on the same spatial grid, but with the background error covariance built from a JONSWAP spectrum for the scene's sea state rather than from a generic smoothness prior, a Gaspari–Cohn localisation radius of 25 m, and an ensemble of 600. This is the arm the headline of Section 5.1 is measured against. It was selected, not designed for the comparison: a pre-registered batch compared three localisation radii and a set of alternative background models and took whichever won, and a later batch that attempted to strengthen the family further did not produce a stronger member. We therefore call it the strongest eligible baseline we possess, and not a strengthened one.

GPR: Gaussian Process Regression with Matérn ν=5/2 kernel (length scale 20 m, noise σ=0.02 m), operating in single-frame mode. No temporal model or physics constraints—pure spatial interpolation. This represents the best achievable single-frame performance without wave physics, providing a reference for the value of temporal tracking and dispersion constraints.

5. Results

Scope of all accuracy results in this section (stated 2026-08-14).. Every accuracy figure reported below was obtained on a **stationary platform** (the episode generator takes no vessel-speed argument and the sensor origin is fixed), with `direction_deg` = 0—following seas, a geometry in which the waves travel *away* from the sensor, which this repository's own tooling labels as the *anti-causal*

configuration for encounter forecasting—and on **synthetic data**. These results are therefore not measured in the encounter geometry of the stated application.

We organize the results around the frozen corpora on which each claim was measured, not around a single dataset. Section 5.1 reports the headline on the crowded stratum, Section 5.2 the non-linear and unseen-seed replication, Section 5.4 the two ablations that attribute the gain, Section 5.5 where the gain sits in space and time, and Section 5.6 the scope, including the measured negative.

5.1. Headline: crowded multi-component seas

On the crowded stratum (episodes with ≥ 15 wave components), the estimator lowers the median normalised RMSE by -38.6% relative to the strongest eligible grid baseline available to us: it is better on 62 of 64 episodes.

What this number is, and is not. **(1)** The corpus and the reference field are synthetic, and that reference is *linear*; the non-linear case is §5.2. **(2)** The learned checkpoint was trained on the same distribution family as the evaluation episodes — seed ranges are disjoint, so "out of sample" here means *on the seed axis only.* **(3)** The baseline is the *strongest we possess*, which is not the same claim as *strengthened until it stopped improving.* **(4)** The sparse stratum is not aggregated into this number and does not support it (§5.6). **(5)** *This batch has no negative control.* The comparator's own skill against a zero field was not measured on this corpus, so the dash in Table 5 is a real absence: we cannot show, for this row, that the margin is not partly the comparator collapsing. **(6)** This batch's gate and analysis scripts were written after its results existed, transcribing rules recorded in its design note; every other row in Table 5 had its analysis frozen first.

Consequently we do not regard this as the paper's strongest evidence. The replication of §5.2 — non-linear reference field, seeds drawn from outside the corpus, comparator verified to retain skill on 47/48 episodes, analysis frozen before results — is stronger on every axis on which the two differ, and it is what

the abstract and the conclusion lead with. This section reports the earlier and larger corpus first because the rest of Section 5 is built on it, not because it is the better measurement, and we report its two weaknesses in the same breath.

5.2. Does it survive a non-linear reference and unseen seeds?

The two most load-bearing confounds in the B-1 result are the linear reference field and the training/evaluation distribution overlap. We removed the first outright and weakened the second by regenerating the corpus with a higher-order spectral (HOS, M=3) reference and seeds drawn from outside the corpus that produced the checkpoint.

The conclusion holds: 43/48 episodes, median relative change -40.2%, $p = 6.84 \times 10^{-9}$ (one-sided sign test). Each batch's preregistration named the contrasts it would test and divided α among them before any result existed; where two contrasts were named, a confirmatory reading required $\alpha/2 = 0.025$. The win counts in Table 5 are compared against thresholds fixed by that division, not selected after the fact.

Critically, this is a comparison *between two arms that both have skill*: the grid baseline itself beats a zero predictor on 47/48 episodes. A margin obtained against a comparator that has itself collapsed is not evidence about the estimator. We therefore print the comparator's own skill in Table 5 wherever the batch measured it. Five of its rows carry a dash instead, because that control was not part of those batches' designs; for those rows the reader has our word about the comparator and not a number.

We do not compare the size of this number with the B-1 margin: the corpora, reference fields, and realised non-linearity grades differ.

5.3. Is the sensor model doing the work?

Replacing the observation model with a physically correct range-dependent raydrop model leaves the conclusion intact (-42.6% over 20 episodes), and both arms beat the zero reference under both models.

5.4. What carries the gain? Two ablations

Baseline comparison alone cannot say *which property* of the estimator is responsible. We therefore hold the deployed pipeline fixed and remove one property at a time.

The learned prior. Replacing the learned set-valued prior with the analytic proposal package, changing nothing else, removes a margin of -36.3% (45/48, $p = 6.56 \times 10^{-11}$). **Disclosure:** the right-hand side of this contrast is *thin* — the analytic package beats a zero predictor on only 15/48 episodes, whereas the learned package does so on 48/48. ⇒ the correct reading is *"the prior is what makes this arm work at all"*, not *"the prior adds a margin on top of a working arm"*.

Continuity of the wavenumber. Snapping atoms to a wavenumber grid and refitting amplitude and phase — leaving every other component of the pipeline untouched — removes a margin of -38.6% (46/48, $p = 4.18 \times 10^{-12}$). The effect is *dose-responsive*: doubling the grid spacing is worse again (47/48). The same ablation reproduces in the sparse stratum (14/16, -67.4%).

What these two magnitudes are not. They are not additive, and they are not shares of any baseline margin. Each was measured on its own corpus against its own comparison arm, so subtracting one from the other — or from the baseline margin — has no defined meaning. In particular, the B-1 margin and the wavenumber-continuity ablation happen to round to the same figure; that is a coincidence of two independent measurements, not evidence that the ablation removes the whole gain. We also checked the grid-snapping ablation separately for confounding with the amplitude-and-phase refit it necessarily performs, and found the contribution of that refit negligible.

⇒ Both components that the title names are supported by an intervention inside our own arm, not only by a comparison against someone else's.

5.5. Where is the gain? Locating it in space and time

A margin measured on a synthetic corpus can be an artefact of *where* the corpus places its observations and *when* it scores them. Both coordinates differ sharply between our synthetic and measured data, so we re-scored the frozen crowded corpus along each.

Space. Decomposing the score by distance to the nearest observation, the gain persists inside the density band that our measured records actually occupy — points within 5 m of an observation — on 45 of 48 episodes (-31.7%), with both arms beating the zero reference there (48/48). Because every other number in this paper is a win count or a relative median, we give the absolute level here as well: on this band the median nRMSE is 0.043 for the deployed path and 0.063 for the comparator, against 0.339 for a zero field. These are medians over the crowded synthetic stratum and should not be read as a level for any other corpus. This is an *exclusion*, not an attribution: we did not run an equivalence test, and aperture, incidence angle, and occlusion remain unmeasured. Not blind.

Time. The synthetic margins are scored at assimilation time; our measured judgements are scored at forecast lead. Re-scoring the same frozen corpus at a forecast rung gives 47/48, -46.3%. **Two disclosures are mandatory here.** First, that rung is at the *lower* end of the measured lead band: the precise statement is *"a rung covered by* 11 *of the* 12 *measured records"*, not *"the same horizon as the measured data"*. Second, **the comparator degrades across the same interval** — the grid baseline beats the zero reference on 47/48 episodes at zero lead but only 34/48 at the reported rung, while our own absolute error *worsens*. ⇒ the growing ratio must not be read as *"we get relatively better at longer lead"*. Not blind.

The two magnitudes in this subsection answer different questions on different re-scorings of the same corpus and have no defined meaning relative to each other: they should not be compared, added, or subtracted, and neither is a decomposition of any baseline margin.

5.6. Scope, stated as measured

The sparse stratum does not separate. Against the deployed analytic package with its ensemble intact, the crowded stratum favours the learned package while the sparse stratum does not; we therefore do not write *"the learned prior wins in every regime"*. (The sparse figure is a re-aggregation of the frozen deployment-verification corpus rather than a pre-registered contrast, and we cite it as such.)

On measured data the learned prior does not reach the decision threshold. Across 12 fixed-station stereo records, and under every arm configuration we tested — including giving the learned representation the same median-of-k field ensemble that the deployed real-data path uses — the learned prior does not reach the pre-registered threshold against a per-record zero-field reference. The ensemble treatment does move the endpoint, and it lowers error by a confirmed margin, but it does not cross the threshold. We report this rather than omit it; a full treatment of the operating boundary, including why the transfer fails, is the subject of a companion thesis.

5.7. Summary of the measurements this paper rests on

Table 5 and Table 7 collect every comparison this paper relies on, and Fig. 3 plots the first of them with exact confidence intervals alongside each comparator's own skill. Each row is one batch, and the number is read out of that batch's frozen verdict file rather than transcribed by hand. For all but one, the decision rule, the comparison, and the reading were fixed in a preregistration before the runs produced a single result row, and the analysis code was frozen by checksum. The exception is the B-1 row: its gate and analysis scripts were written after the results existed, transcribing rules that had already been recorded in the batch's design note. We state this rather than let the word "pre-registered" cover it. Table 5 is what the representation earns on synthetic corpora, where ground truth exists; Table 7 is where it stops, on the 12 measured records. Fig. 2 shows one reconstruction in full, as an illustration rather than as evidence.

Table 5: What the representation earns. All contrasts are paired per episode and one-sided in the pre-registered direction; "better" counts episodes on which the first arm has lower nRMSE, and the relative median is the median of the paired relative differences. Crowded stratum unless noted. *Read the third column with the fourth*: a large margin against a comparator that itself has little skill is not the same measurement as a large margin against one that does, which is why the "cmp." column is printed rather than described: it counts the episodes on which the *comparator* beats a zero predictor, i.e. how much skill the thing we beat had, and a dash means that control was not part of that batch's design. These are separate experiments on different corpora and their magnitudes have no defined meaning when added or subtracted — they should not be combined arithmetically.

Question	Batch	n	better	rel. median	p	cmp.
Against an external baseline						
Strongest eligible baseline	B-1	64	62	-38.6%	1.13×10^{-16}	—
Non-linear reference, fresh seeds	B-12	48	43	-40.2%	6.84×10^{-9}	47/48
Physically correct sensor model	B-15	20	17	-42.6%	1.29×10^{-3}	20/20
Sea state matched to records	B-26	24	24	-42.4%	—	23/24
Ablations of the representation itself						
Continuity of the wavenumber	B-16	48	46	-38.6%	4.18×10^{-12}	—
— sparse stratum	B-19	16	14	-67.4%	2.09×10^{-3}	—
The learned prior	B-22	48	45	-36.3%	6.56×10^{-11}	15/48
Ensemble on learned path	S-ENS	64	55	-17.9%	1.77×10^{-9}	—
— sparse stratum	S-ENS	64	35	*undecided*	—	—
Where the gain sits						
In space (inside observed band)	M-10	48	45	-31.7%	6.56×10^{-11}	48/48
In time (lead $1.0\,T_p$)	T-4	48	47	-46.3%	—	34/48

Table 6: Absolute accuracy, in the units the measurement is made in and in metres. The synthetic harness normalises by significant wave height, so metres follow from RMSE = nRMSE $\times$ H_s with no new measurement; this table is a re-aggregation of the same frozen rows that produce Table 5. Each cell is the median over episodes of the per-episode median over estimator seeds, with the interquartile range. *Read within a sea state, not across:* H_s differs between the two, so the ordering in metres and the ordering in nRMSE need not agree across rows.

Batch	Arm	sea state	H_s [m]	n	nRMSE	RMSE [m]
B-12 non-linear reference	`SHIP-NOW`	BF4	1.2	24	0.137 [0.096,0.203]	0.164 [0.115,0.244]
	`SHIP-NOW`	BF5	2.0	24	0.130 [0.096,0.158]	0.259 [0.191,0.315]
	`SPEC25`	BF4	1.2	24	0.209 [0.201,0.218]	0.251 [0.241,0.261]
	`SPEC25`	BF5	2.0	24	0.199 [0.191,0.213]	0.399 [0.381,0.425]
B-16 wavenumber ablation	`KGRID1`	BF4	1.2	24	0.228 [0.202,0.244]	0.273 [0.243,0.292]
	`KGRID1`	BF5	2.0	24	0.204 [0.176,0.228]	0.408 [0.353,0.457]
	`KGRID2`	BF4	1.2	24	0.302 [0.278,0.316]	0.363 [0.334,0.379]
	`KGRID2`	BF5	2.0	24	0.278 [0.247,0.318]	0.556 [0.494,0.635]
	`SHIP-NOW`	BF4	1.2	24	0.137 [0.096,0.203]	0.164 [0.115,0.244]
	`SHIP-NOW`	BF5	2.0	24	0.130 [0.096,0.158]	0.259 [0.191,0.315]

What each n is made of, and why it is not n independent trials. The episode counts in Table 5 are not independent draws, and we state what they are. The B-1 corpus of 64 crowded episodes is a full factorial: 16 generator seeds crossed with two Beaufort levels and two component counts. The 48-episode corpus used by the ablations and by the non-linear replication is the same design with 12 seeds. So four episodes share each seed, and the true number of independent scene draws behind an n of 48 is 12, not 48.

The sign tests we report treat episodes as exchangeable. That is the convention the preregistrations fixed, and we keep it rather than change the rule after the fact, but it means the p-values should be read as descriptive of the paired pattern rather than as calibrated tail probabilities: if the seed is the dominant source of scene-to-scene variation, the effective sample size lies somewhere between 12 and 48, and every p we print is correspondingly optimistic. No result in this paper turns on a marginal p — the confirmatory rows clear their thresholds by several episodes — but a reader comparing two of them on the strength of p alone would be over-reading.

How the rows of Table 5 must be read. Four qualifications apply to the table as a whole and are not repeatable per row. *(i)* Every corpus in it is synthetic, and all but the B-12 row use a linear ground-truth field; the gains are synthetic gains, and the non-linear check is the B-12 row alone. *(ii)* The learned checkpoint was trained on scenes from the same distribution family as these corpora. Only the seed axis is out of sample: B-12 adds unseen seeds, and no row tests a shift of the distribution itself. *(iii)* The comparator in the first block is the strongest eligible grid baseline we possess, which is not the same claim as a baseline strengthened for this comparison; the "cmp." column is there so the reader can judge how much skill that comparator actually had in each corpus, which is the negative control for every row that has one. Within the first block, the B-1 row is the one whose comparator skill was not measured on this corpus, and it is the only row anywhere in the table whose gate and analysis scripts were written after its results existed; Section 5.1 says both where that number is first

reported. *(iv)* The B-16 and B-19 magnitudes are not attributable by subtraction from anything else in the table: a separate batch checked that experiment for confounding and found it negligible, which is what licenses reading them as sizes rather than as directions. The B-26 and M-10 rows were not blind; the others were analysed with the decision rule frozen before any result row existed.

Two rows of Table 5 carry a caveat that the table itself makes visible, and we state it rather than let the reader find it. In the B-26 row the margin is the largest in the table, but the comparator degrades in that corner: its own margin over a zero predictor falls to -4.0% there from -16.2% in the anchor corner, and our own absolute error is *worse* in the corner with the larger margin (0.137 against 0.118). The ranking of the four corners therefore reverses between the ratio scale and the absolute scale, and no claim about which sea state suits the method best is available from this experiment; what is available is the exclusion statement that the boundary does not lie along the sea-state axis, on a linear-ground-truth corpus. Two further facts belong with this row. The baseline's localisation radius was held at the same 25 m in all four corners, a value chosen on the long-period corners; in the shortest corner the peak wavelength is 19.0 m, so the ratio of localisation radius to wavelength moves from 0.33 to 1.32 across the table. That was an undisclosed confound when the batch was run, and it was closed afterwards by intervention rather than by argument: re-deriving the radius downward, preserving the ratio, made the baseline *worse* or left it unchanged, so 25 m was already at or below its optimum. And the comparator in the two short-period corners passes a sign test but does not clear the 5% skill floor that our own baseline tuning protocol requires against a zero predictor. Both facts point the same way: the margin in those corners is partly the comparator's weakness, which is why we print the "cmp." column. The T-4 row has the same shape: the comparator falls from 47/48 at zero lead to 34/48 at the pre-registered rung, so the growth of the margin with lead time is not evidence that the method improves with lead time. The T-4 rung is at the lower end of the measured lead band, 11 of the 12 records, not at the horizon the real-data section reports.

Table 7: Where it stops. Every row is a measurement on the 12 fixed-station stereo records, scored against the per-record true zero-field reference. "beats" counts records on which the arm has lower error than that reference; the pre-registered decision threshold for a confirmatory reading was 11/12. *None of these rows is a comparison between arms* — each asks only whether the arm has any skill at all.

What was varied	Batch	beats	reading
Noise model (range- and angle-dependent)	B-17	4/12	no skill
Observation density and noise, best corner	B-18	8/12	below threshold
Noise *shape*, best of four models	B-14b	8/12	below threshold

Table 7 is the paper's central negative and the reason its title speaks of an operating boundary. Four candidate explanations for it were measured and none accounts for it: the shape of the noise (B-14b), the density of the observations and the noise level together (B-18), a realistic range- and angle-dependent noise model (B-17), and the sea state (B-26, in Table 5). The learned prior does not reach the pre-registered decision threshold on the measured records under any arm configuration we ran, and we report that as the measurement it is rather than as a limitation to be addressed later.

The two S-ENS rows of Table 5 deserve a word, because they are the only rows there that do not describe the shipped system. They compare the learned initialiser drawing three atom sets and taking the median against the same initialiser drawing one, which is the shipped default (1). The crowded stratum favours the ensemble on 55 of 64 episodes against a pre-registered threshold of 40; the sparse stratum is undecided, and the smallest win rate that design could resolve was 0.682. We report it because the same operator earns on the measured records, and whether that transfers is otherwise an open question — *not* as a recommendation to change the default, which would need its own preregistration and a cost measurement we do not have.

One positive does exist on the measured records and it is not a skill claim: giving the proposal an ensemble lowers its forecast error at the 1–3 s horizon (28/36, -8.5%, $p = 5.97 \times 10^{-4}$), and the same operator gives 35/36 (-11.1%)

on the far band; this is a within-arm contrast, and the endpoint against a zero predictor remains undetected. Four things belong with that number. The unit is not the record: 36 counts record–lead cells, 12 records at three lead bands, and the record-level statement is weaker. The analysis was not blind, and one of the two bands was partly determined by an answer we had already seen. The smallest win rate the design could resolve was 0.742. And the 12 records come from four fixed stations, so nothing here generalises to the site level.

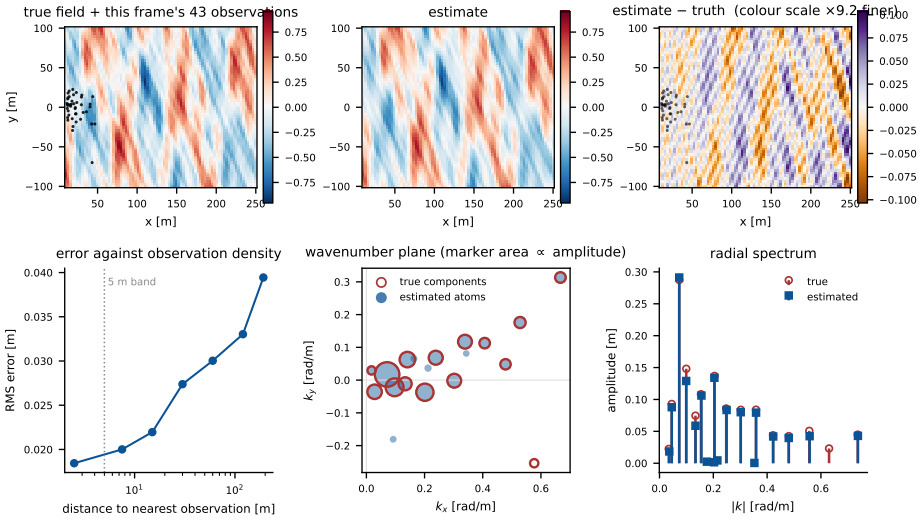


Figure 2: One crowded episode, one frame, illustrative only — *no comparison is drawn here*, and every quantitative claim in this paper lives in Table 5. Top: the true field with that frame's observations overlaid, the reconstruction, and their difference on a ten-times finer colour scale. The observations occupy a small patch near the sensor; the field is reconstructed across the whole domain from the spectral parameters, which is what "grid-free" buys and also where it can go wrong. Bottom left: RMS error against distance to the nearest observation, which roughly doubles from the observed patch to the far corner — the error is not uniform, and a reader should not read a single domain-average number as a local accuracy. Bottom centre and right: the estimated atoms against the true components. Most are recovered; one true component near $(k_x, k_y) = (0.57, -0.25)$ rad/m has no estimated atom near it, and one estimated atom sits where no true component does. The provenance of this figure — episode, frame, seed, configuration, and commit — is recorded alongside it.

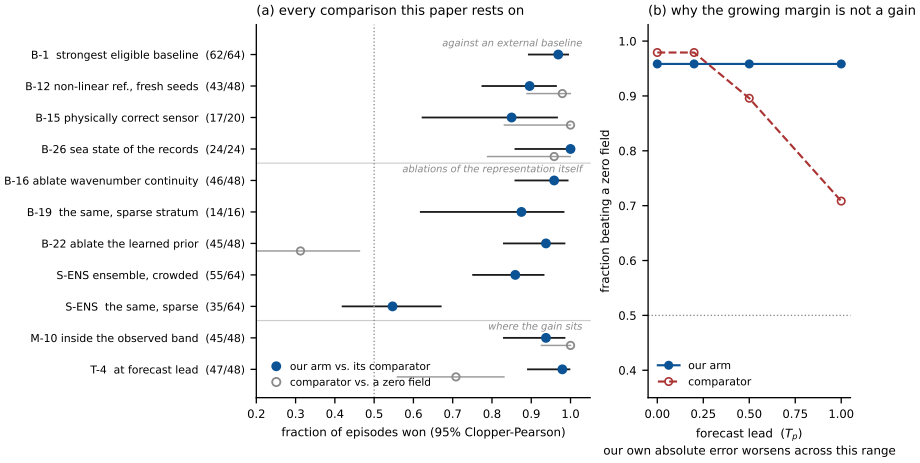


Figure 3: (a) Every comparison in Table 5, as a win fraction with an exact 95% Clopper–Pearson interval. Filled markers are our arm against the comparator that batch used; hollow markers are that comparator against a zero field, i.e. how much skill it had. Rows without a hollow marker are the ones whose batches did not include that control. The sparse S-ENS row is the one contrast here that does not separate. (b) Why the margin that grows with forecast lead is not a gain: our arm's own skill against a zero field is flat across the ladder, while the comparator's falls away, so the ratio widens because the denominator weakens. Generated from the frozen verdict files by `scripts/gen_fig_results.py`.

A spectral-domain metric, and what it took to get one. Everything above is one metric. We add one spectral-domain measurement, and the route to it is part of the result. Our first attempt was pre-registered and *failed its own negative control*: that design required a zero field to score below *both* arms on the two-dimensional wavenumber-spectrum R^2, and on a substantial minority of episodes the analytic comparator scored below a zero field instead — placing energy at the wrong wavenumbers is penalised by that metric while placing none is not. Its preregistration made that disqualifying, so it produced no label and we do not quote its contrasts. Relaxing a gate after seeing what it caught is the failure mode this protocol exists to prevent.

We therefore re-ran the measurement on a *different* corpus — disjoint seeds and a linear rather than a higher-order reference — with the control split into three: does the metric behave, does *our* arm beat a zero field, and how often does the comparator. The first two block; the third is reported. On 32 crowded episodes the deployed learned initialisation matches the reference wavenumber spectrum better than the analytic one on 25 ($p = 1.05 \times 10^{-3}$). *We grade this exploratory and not confirmatory*, because the first attempt's numbers were known when the second was designed.

Three things travel with it. The comparator beats a zero field on only 21 of the 32 episodes, so this margin is partly the comparator's weakness and its size should not be read as an accuracy ratio. Our own arm is below a zero field on two episodes. And the metric compares power spectra, which for a real field are symmetric under $\mathbf{k} \to -\mathbf{k}$: it does not distinguish propagation direction, so it is a spectral-magnitude agreement measure and not a directional one. The companion amplitude-recovery contrast did not reach its threshold (20/32) and we draw no conclusion from it.

What was removed, and what that costs. The three subsections that previously stood here — an ablation of the dispersion constraint and the confidence weighting, a raydrop-model sensitivity study, and a sparsity/noise/sea-state robustness study — together with four tables and four figures, have been moved to the sup-

plementary archive rather than deleted. Their corpus is the one whose headline comparison is retracted, and two of them were additionally produced before a correction to the tracker's phase advance. Where a pre-registered replacement exists it appears in the two tables above. Where none exists, the quantity is simply absent from this paper and we do not restate it from the archived run: this applies to the five-metric evaluation panel, the stratification against wave buoys, the twelve-episode comparison against tuned grid baselines, the spectral-accuracy comparison, and the absolute per-frame latency, which has never been measured on the deployed learned initialisation path at all. The paper is thinner for this, and a reader comparing it against the earlier version will find fewer numbers rather than more. That is the intended direction: every number that remains is one a frozen verdict file will reproduce.

6. Discussion

6.1. Why a grid-free representation, and where it stops helping

A grid state must resolve the field everywhere it is defined, so its cost and its conditioning are set by the resolution rather than by the complexity of the sea. A spectral state pays instead for the number of components it carries. That is the structural argument for the representation, and the ablations in Section 5.4 are what turn it from an argument into a measurement: removing only the continuity of the wavenumber — keeping every other component of the pipeline — removes a margin of -38.6%, and the loss grows when the grid is coarsened. We checked separately that this ablation is not confounded with the amplitude-and-phase refit it necessarily performs, and the magnitudes here are *not additive* with the baseline margin: they were measured on different corpora against different comparison arms.

Where it stops helping, stated as measured. The advantage is not uniform across regimes, and we no longer characterise its regime dependence: the earlier characterisation was computed on a corpus in which the compared arms were never scored on the same frames, and we withdraw it rather than restate it. What

we can say from the present corpora is narrower and is already in Section 5.6: the margin is measured on the crowded stratum; against the deployed analytic package the sparse stratum does not separate; and on measured records the representation does not reach the decision threshold at all.

6.2. Effect of observation model

Two questions about the observation model have been measured on frozen batches rather than argued. First, whether the geometric return-sampling model changes the conclusion: it does not, and the physically correct slope model is in fact the more favourable of the two (B-15, Table 5), with the comparator retaining skill in both. Second, whether the *shape* of the observation noise explains the negative on the measured records: it does not, and the best of four noise models still reaches only 8/12 (B-14b, Table 7). That fraction counts records on which the arm beats the per-record true zero-field reference; it is not a comparison between arms, and asks only whether the arm has any skill at all. The earlier cross-arm sensitivity figures are withdrawn with the corpus they came from and are in the supplementary archive.

6.3. Hyperparameter sensitivity

The four hyperparameters of the analytic path (lr=0.01, λ_1=10^{-4}, τ_{prune}=0.10, τ_{grow}=0.05) were selected by a single coarse sweep during development and then held fixed across every batch reported here — across Beaufort 2–5, 1–30 components, several ground-truth classes, and both strata. That one configuration was never re-tuned per experiment is a property of the protocol and can be checked in the preregistrations; it is the only sensitivity statement we make. The development-time perturbation sweep that the earlier version of this section reported is not restated: it was run on the retracted corpus and, for λ_1, at values that predate the change of the shipped default to 10^{-4}. It is in the supplementary archive.

6.4. Practical deployment considerations

WaveSplat's computational footprint on a single CPU thread (measured in Section 3.5; an earlier figure quoted here was withdrawn, Appendix Appendix A, R3) is intended for deployment on standard shipboard computing hardware. Inference does not require GPU acceleration for the tracking phase, and no offline calibration against a particular site is needed. It does require the learned proposal checkpoint at initialisation, which is where the GPU is used when one is available. The spectral parameters are directly interpretable: a naval architect can inspect the estimated wavenumber vector, amplitude, and frequency of each component, enabling sanity checking against visual observation and meteorological forecasts.

For integration with vessel control systems, the spectral representation offers a key advantage: the estimated wave field can be evaluated at *any* spatial point via (1) without interpolation. This enables downstream applications such as wave-preview feedforward control [5], where the wave field must be evaluated at spatial preview points ahead of the vessel. Grid-based methods would require interpolation between grid points, introducing additional error and latency.

6.5. Temporal extrapolation as validation of spectral fidelity

Because the estimated parameters satisfy the dispersion relation by construction, the forward model (1) can be evaluated at any future time without re-estimation. Whether that extrapolation is accurate is therefore a test of the estimated (k_x, k_y, ω, ϕ) rather than a forecasting contribution, and it is reported in Section 5.5 as the T-4 row of Table 5: at a lead of $1.0\,T_p$ the deployed path is ahead on 47 of 48 episodes. That row must be read with its negative control, because the comparator's own skill falls from 47/48 at zero lead to 34/48 at that rung; the rung sits at the lower end of the measured lead band, 11 of the 12 records. The earlier five-episode horizon table, produced before the tracker phase-advance correction, is in the supplementary archive.

6.6. Limitations and future work

Several limitations should be acknowledged.

Online model order does not contribute accuracy. The pruning-and-growth mechanism of the retracted corpus (archived) does select scene-matched component counts, but under the corrected tracker its accuracy contribution was null on the twelve episodes it was measured on, against a seed-to-seed noise floor far larger than the effect, and it was actively harmful on most of the crowded ones, where the fixed energy criterion prunes to fewer components than the scene contains. We accordingly present it as a mechanism and a compute-efficiency device rather than as a contribution. A model-order criterion that is aware of crowding—for instance, energy retention conditioned on the beamforming main-lobe width rather than a fixed 90% quantile—is the natural repair and is untested.

Directional spread of the synthetic distribution. Compared against a large buoy archive, our generator occupies the lower tail of the observed directional-spread distribution and its peak-period ceiling excludes swell, so the synthetic universe is narrower and shorter-period than the measured one. The per-percentile figures are not restated here: they come from the retracted corpus and have not been re-measured. The repair belongs to the generator, not the estimator—directional spreading functions and a T_p range extending into swell—and we deliberately did not change it mid-study so that the synthetic results remain from one frozen dataset. We do not characterise which regime the advantage is largest in; that characterisation was withdrawn in Section 6.1.

The positive results are synthetic. Every margin reported in Section 5.1 and its ablations is measured on simulated fields with a simulated sensor. Measured data enters this paper only through the negative in Section 5.6: 12 fixed-station stereo records on which the learned prior does not reach the decision threshold. Real LiDAR from a moving platform may contain systematic biases (multi-path, spray) that neither our noise model nor those fixed-station records represent, and no shipboard measurement of any kind supports this paper — the platform term was removed from the title for that reason. We note that the simulator and estimator are covered by an automated test suite (263 tests at the time of

writing; the count is generated, not typed) (measured 2026-08-13; a further 8 shipping-default regression tests were added subsequently) covering dispersion, Gerstner kinematics, Fresnel optics, and raydrop statistics — note that the Gerstner test exercises the kinematics *function*, which the generation pipeline does not use for horizontal displacement, so it does not certify that the generated data is nonlinear. What the protocol does provide instead is narrower and checkable: each comparison in Section 5 is paired per episode, its direction and decision threshold fixed before results existed, its analysis code checksum-frozen (with the one exception noted there), and its number read from the resulting verdict file rather than transcribed. Like the LiDAR-based wave estimation studies that preceded us, we establish the algorithm in simulation before any tank trial [13, 3]; unlike a tank trial, simulation cannot tell us whether the result transfers, which is what Table 7 is for.

Linear forward model. Earlier versions stated that the $O(\epsilon^2)$ model mismatch was small but non-zero and affected all methods equally (Appendix Appendix A, R2). The correct facts are: the synthetic ground truth is a *displacement-free linear superposition on a fixed grid*, so it lies in the same function space as the estimator's model equation and the measured mismatch is *zero*; and a zero mismatch is not neutral between methods, because the grid-based baselines do not use that representation. The linear approximation does degrade for steep waves, and a second-order Stokes extension would preserve the spectral parameterisation. *Non-linear validation was performed in a separate higher-order-spectral (M=3) batch, not on this corpus.*

Tracking phase dependence on initialization. The `w_phi` tracking mode freezes wavenumbers and updates only amplitudes and phases, so components missed during beamforming initialization cannot be recovered during tracking. On the originally released dataset this produced tracking nRMSE 2–3× higher than the post-initialization nRMSE whenever high-wavenumber components were missed, and we previously described the cutoff as a beamforming detection threshold at $|k| \gtrsim 0.4$ rad/m. That description was incorrect. A mesh-independent test of

the initializer alone—synthesising LiDAR samples directly from analytic components, bypassing the raycast mesh, with the real sensor sampling geometry—finds no such threshold: the beamformer recovers isolated components across $|k| = 0.15$–1.00 rad/m and down to amplitudes of 0.01 m. The value 0.4 rad/m corresponds to $\lambda \approx 15.7$ m, which is where amplitude transmission through the original rendering mesh collapses (Section 4.2); it was a property of the reference field, not of the algorithm. What remains true, and is the point of the limitation, is the structural one: because `w_phi` freezes wavenumbers, whatever the initializer misses is unrecoverable thereafter, so tracking quality is capped by initialization quality regardless of why a component was missed. The net benefit of tracking is therefore concentrated in providing real-time computational efficiency ($\sim$10 gradient steps vs. $\sim$200 for full re-initialization) rather than accuracy improvement beyond initialization. The corrected tracker sharpens this into a quantitative statement. Comparing per-episode post-initialization against post-tracking nRMSE across the 24 arm–episode pairs of the re-measured ablation and model-order experiments, tracking ends *above* its own initialization on most of the arm–episode pairs it was measured on, but by a small and bounded margin. So the honest characterisation of Phase 2 on this dataset is not that it degrades the estimate—the earlier 2–3$\times$ inflation and the $0.009 \rightarrow 0.08$ collapse on the calmest episodes were properties of the uncorrected tracker and do not reproduce—but that it holds the initialization roughly in place at a fraction of the per-frame cost. Tracking is a compute mechanism, not an accuracy mechanism.

Residual amplitude under-estimation. The estimator recovers less variance than the true field, and on the retracted corpus that deficit did not differ between the sparse and crowded halves even though nRMSE did. The levels are not restated here: variance ratio was part of the secondary panel that was not re-measured for the batches reported in Section 5. We previously attributed this deficit to Adam momentum accumulating oscillating gradients from an inaccurate phase prediction $\phi_i \mathrel{+}= \omega_i \Delta t$. That attribution is withdrawn: the specific inaccuracy

it named was a double application of the phase advance, which has since been corrected, and the deficit narrowed from 0.66 on the original mesh through 0.83 to the present 0.916 without the momentum path being touched. The remaining 8% is therefore *unattributed*—we can bound it but not currently explain it, and we prefer to say so rather than carry forward a mechanism whose named cause no longer exists. Post-hoc least-squares amplitude calibration is effective when dense observations are available but does not improve with sparse LiDAR (∼50 pts/frame) due to insufficient spatial coverage; isolating the residual mechanism (momentum-free optimizers for amplitude, periodic optimizer state resets, or a per-frame amplitude least-squares solve) is future work.

Initialization resolution and the peak-separation radius. The greedy beamforming peak search enforces a fixed minimum separation $\Delta k_{\min} = 0.15$ rad/m between accepted peaks, so component pairs closer than this in wavenumber space are structurally unresolvable by our initializer regardless of signal-to-noise ratio. This is not a corner case: a majority of the simulated episodes—and every 15-, 20- and 30-component episode—contain at least one true pair separated by less than 0.15 rad/m. (The exact count came from the retracted corpus and is not restated.) A pre-registered sweep of $\Delta k_{\min}$ (5 seeds per cell, decision rule fixed in advance, with a control episode containing no close pair) shows the constant to be a regime-dependent trade rather than a universal floor: relaxing it to 0.05 rad/m improved the crowded 15-component case on every seed but degraded the 5-component case on every seed, because a smaller radius lets the greedy search stack duplicate peaks on a single true component; the control episode was unaffected. The magnitudes are not restated: that sweep is from the retracted corpus and has not been re-measured. Two consequences follow. First, the crowded-regime error is *not* explained away by this constant: the bulk of it survives when the constant is removed. Second, statements in this paper of the form "these components are not recovered" are properties of the estimator as configured, and understate what this estimator class can achieve in crowded seas by a few percent; they should not be read as method-independent impossibility.

The principled repair is not a smaller fixed radius but an adaptive one, scaled to the beamforming main-lobe width and hence to the effective aperture; we have not tested it. For $N_{\text{true}} \geq 30$, wavenumber-space crowding additionally exceeds what the beamforming grid resolves at any separation radius; higher-resolution or 3D (k_x, k_y, ω) FFT could help.

Current estimation. We do not estimate ocean currents, whose Doppler shift makes the apparent dispersion $\omega = \sqrt{g|k|\tanh(|k|d)} + \mathbf{k} \cdot \mathbf{U}$; a 0.5 m/s current introduces $\sim$5% frequency bias at BF4. Joint current–wave estimation is a natural extension.

Cold start and re-initialization. Phase 1 requires $T_{\text{init}} = 2T_p$ of observations (e.g., $\sim$16 s at BF4) before tracking begins; during this interval no wave estimate is available. If the sea state changes abruptly (e.g., vessel course change exposing a new swell system), re-initialization is needed, incurring the same latency. A background re-initialization scheme running in parallel with tracking could mitigate this in practice.

Surface normal approximation for real data. In simulation we use the exact surface normal of the reference field for confidence weighting (that field is a displacement-free linear superposition, not a Gerstner surface; see Sections 3.2 and 4) c_j, but real-data application must substitute the flat-surface approximation $\mathbf{n} \approx \hat{\mathbf{z}}$. The resulting bias scales with wave slope (up to $\sim$9% in $\cos\theta_j$ at grazing incidence at BF4); wave-tank validation is planned to characterise the end-to-end nRMSE impact.

Observation weight nomenclature. The weighting $c_j = \cos\theta_j/(r_j/d_{\text{ref}})$ is a geometry-aware simplification motivated by the incidence-angle dependence of specular reflection and the inverse-square range attenuation. The exact Fresnel reflectance $R(\theta)$ [35] is not used directly, as it would additionally require surface-roughness and polarisation assumptions that are difficult to justify from observation alone. The name "geometry-aware" (previously "Fresnel-aware" in an earlier draft) more accurately reflects the functional form.

Baseline fairness. We cannot claim to have beaten a fully tuned ensemble filter. The sweep and cross-baseline comparisons that once supported such a claim were run on the retracted corpus, and the one pre-registered attempt to re-establish it against a repaired baseline returned an undecided primary. What survives is the narrower statement of Section 5.1: the comparator there is the strongest eligible baseline we possess, not a baseline strengthened for the occasion. Further tuning of ensemble filters, or hybrid ensemble–spectral approaches, remain unexplored.

Uncertainty quantification. WaveSplat returns point estimates; Hessian-based confidence intervals or multi-start ensembles could provide uncertainty.

Directional spreading. Each component is monochromatic; representing continuous directional distributions (e.g., $\cos^{2s}$) would require directional or von Mises basis functions.

Ship motion. The current simulation assumes a stationary sensor platform. In practice, heave, pitch, and roll of the vessel change the LiDAR beam pattern frame-to-frame; an IMU/INS-based coordinate transform is a necessary preprocessing step for sea-going deployment, and the residual motion-compensation error constitutes an additional noise source not modelled here.

Near-surface false returns. At 1550 nm the absorption length of sea water is approximately 1 mm [36], so essentially no light penetrates below the surface; sub-surface volume scattering is negligible. However, sea spray, whitecaps, and rain droplets above the nominal water surface can generate false LiDAR returns that are indistinguishable from surface elevation measurements. These super-surface returns introduce a positive elevation bias not captured by the additive noise model, and their prevalence increases with sea state; empirical calibration on real data is needed to assess their impact.

Relationship to physics-informed learning. WaveSplat can be viewed as physics-constrained optimization with a hand-designed plane-wave basis, structurally analogous to PINNs that use a neural-network basis. The key advantage of the

explicit plane-wave basis is threefold: (i) the dispersion relation is satisfied by construction rather than as a soft penalty, (ii) beamforming provides an efficient spectral initialization that neural bases lack, and (iii) the $4N$ parameter count is orders of magnitude smaller than typical PINN architectures, reducing overfitting risk with sparse observations. Hybrid approaches—e.g., learned initialization or learned observation weighting—could combine the interpretability of spectral components with the flexibility of neural representations.

7. Conclusion

We estimate phase-resolved ocean wave fields from sparse 3D point clouds as a small set of continuous-wavenumber spectral atoms under a dispersion constraint, with the initial configuration proposed by a learned set-valued prior. On crowded multi-component seas with a non-linear reference field and out-of-corpus seeds, this lowers median normalised RMSE by -40.2% against the strongest eligible grid baseline we possess (43/48 episodes), in a comparison where that baseline itself retains skill against a zero predictor (47/48).

What we regard as the substantive contribution is not the margin but the *attribution*. Holding the deployed pipeline fixed and removing one property at a time, the learned prior accounts for -36.3% (45/48) — measured against an analytic package that itself clears a zero predictor on only 15/48 episodes — and the continuity of the wavenumber accounts for -38.6% (46/48, dose-responsive, and reproducing in the sparse stratum at 14/16). The gain also survives re-scoring inside the observation-density band our measured records occupy (45/48) and at a forecast rung that 11 of those 12 records cover (47/48) — though in the latter case the comparator degrades across the same interval, so the growing ratio must not be read as a growing advantage. Neither re-scoring was blind — the expected direction was known before the runs, so they are not blind replications — and that rung sits at the lower end of the measured lead band rather than spanning it. The two ablation magnitudes are *not additive* with each other or with the baseline margin — different corpora,

different comparison arms — and we verified separately that the grid-snapping ablation is not confounded with the refit it necessarily performs.

What we did not obtain. On 12 fixed-station stereo records the learned prior does not reach the pre-registered threshold against a per-record zero-field reference under any arm configuration we tested, including one in which it is given the same median-of-k field ensemble that the deployed real-data path uses. That treatment does lower error by a confirmed margin and does move the endpoint, but not across the threshold. We state this as a measured boundary rather than as an omission: the representation that earns its gain on crowded synthetic seas has not been shown to earn it on the measured data we hold, and locating that boundary — rather than widening the margin — is where we believe the remaining work lies.

We also note two limits of the evidence itself. The synthetic gains are confined to the crowded stratum; against the deployed analytic package the sparse stratum does not separate. And the learned checkpoint was trained on the same distribution family as the synthetic evaluation, so "out of sample" on that corpus means on the seed axis only.

CRediT authorship contribution statement

The sole author is responsible for all aspects of this work.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Changes from earlier versions

Earlier versions of this manuscript made four claims that later measurement withdrew. We list them here rather than in the running text, so that the body

Table A.8: Withdrawals and corrections, with what replaced them.

	Withdrawn	Why, and what stands now
R1	Geometry-aware confidence weighting as a *contribution* (2026-08-14)	A fresh out-of-sample corpus found disabling the weighting better in both strata. The mechanism is still deployed and still described, because a later ablation found that *removing* the ground-truth-derived weighting also hurts; what does not stand is calling it a contribution of this work.
R2	"The ground truth is Gerstner (non-linear), so the $O(\epsilon^2)$ mismatch affects all methods equally" (2026-08-14)	False on both halves. The generator discards the horizontal displacement, so the reference field is a displacement-free linear superposition and the measured mismatch against the model equation is exactly zero. The non-linear case is now measured separately with a higher-order reference (Section 5.2).
R3	All absolute per-frame latencies (2026-08-14)	They came from separate processes without thread pinning, so they measured the harness as much as the estimator. *Superseded*: the deployed path has since been timed in one process on a single thread and is reported in Section 3.5. No baseline arm was timed, so no comparative timing claim replaces the withdrawn ones.
R4	"The dispersion constraint contributes accuracy" (2026-08-14)	The ablation that supported it removed the frequency initialisation together with the constraint, so it did not isolate either. What stands is the parameter-count reduction, which is structural. A controlled intervention has not been run.

states only what is currently supported; each is referenced from the section where the claim used to appear.

A fifth class of change is not listed above because it is not a withdrawal of a claim but of a *corpus*: an earlier evaluation corpus was found to score the compared arms on different frames, and every cross-arm number computed on it has been removed from this paper. Section 5.7 states which quantities had a pre-registered replacement and which are simply absent as a result.

Data availability

Code, simulation data, experiment scripts, *and the trained prior checkpoint* will be deposited on a public repository upon acceptance. Every result in Section 5 carries a batch identifier (B-1, B-12, ...); each names a directory in that repository holding its preregistration, its checksum-frozen analysis code, its raw result rows, and the verdict file from which this paper's numbers are read. The "supplementary archive" referred to in Sections 5.7 and 6 is a single file in that repository holding the subsections, tables, and figures withdrawn from this manuscript, kept verbatim with the reason for each withdrawal, so that a reader can see what was removed rather than only that something was. The complete 100-episode dataset can be regenerated from the released simulator with the provided configuration files and fixed random seeds, ensuring end-to-end reproducibility of the tables and figures that were computed on it. The headline results of this paper were not: they come from the frozen corpora of the batches identified in Table 5.

The buoy directional-spread statistics used for the external-validity stratification of the retracted corpus (archived) are third-party observations from the Coastal Data Information Program [37], retrieved over OPeNDAP on 2026-08-03. We release the station list, the pre-fixed sampling rule, the retrieval and estimator scripts, and the derived per-record spread values, so that the stratification is reproducible without redistributing CDIP data.

We could not locate a public, phase-resolved, two-dimensional *lidar* sea-

surface dataset, and we state the scope of that search rather than assert non-existence. The measured evaluation in this paper therefore uses stereo reconstructions from fixed stations rather than lidar, which is itself a scope limit: the sensor that motivates the method is not the sensor the negative result was obtained with. We enumerated the collections of the USACE Field Research Facility THREDDS server to collection level—its `frf/` and `projects/` trees and the nine collections at the server root—and found one candidate, a 30-file, three-day scanning-lidar wave-shape deployment; that campaign's lidar coverage is three days of a sixteen-day campaign, which is a property of the deployment rather than of our search. We also confirmed that no public download path exists for the processed free-ocean wave dataset (FOWD) product. Airborne scanning-lidar measurements of open-ocean waves do exist in the literature [12]; what we could not find is a public archive of the underlying point data. This is a statement about the reach of our search, and a dataset outside it would not surprise us.

Acknowledgments

The author thanks the Avikus engineering team for discussions on shipboard LiDAR integration and practical deployment considerations.

References

[1] O. M. Faltinsen, Hydrodynamics of High-Speed Marine Vehicles, Cambridge University Press, 2005.

[2] L. H. Holthuijsen, Waves in Oceanic and Coastal Waters, Cambridge University Press, 2007.

[3] N. Desmars, M. Hartmann, J. Behrendt, N. Hoffmann, M. Klein, Nonlinear deterministic reconstruction and prediction of remotely measured ocean surface waves, Journal of Fluid Mechanics 975 (2023) A8.

[4] R. Vettor, C. Guedes Soares, Development of a ship weather routing system, Ocean Engineering 123 (2016) 1–14.

[5] S. Park, Wave-preview feedforward control for hydrofoil vessels using LiDAR-based spectral wave estimation, Ocean Engineering (2026). Submitted.

[6] Det Norske Veritas, Recommended Practice DNV-RP-C205: Environmental Conditions and Environmental Loads, Technical Report, DNV, 2010.

[7] Datawell BV, Datawell Waverider Reference Manual, 2009.

[8] J. C. Nieto-Borge, G. R. Rodríguez, K. Hessner, P. I. González, Inversion of marine radar images for surface wave analysis, Journal of Atmospheric and Oceanic Technology 21 (2004) 1291–1300.

[9] I. R. Young, W. Rosenthal, F. Ziemer, A three-dimensional analysis of marine radar images for the determination of ocean wave directionality and surface currents, Journal of Geophysical Research: Oceans 90 (1985) 1049–1059.

[10] P. Queffeulou, Long-term validation of wave height measurements from altimeters, Marine Geodesy 27 (2004) 495–510.

[11] M. R. Belmont, J. M. K. Horwood, R. W. F. Thurley, J. Baker, Shallow angle wave profiling LIDAR, in: 2008 IEEE/OES 9th Working Conference on Current Measurement Technology, 2008, pp. 217–223. doi:10.1109/CCM.2008.4480871.

[12] P. A. Hwang, W. B. Krabill, W. Wright, R. N. Swift, E. J. Walsh, Airborne scanning lidar measurement of ocean waves, Remote Sensing of Environment 73 (2000) 236–246. doi:10.1016/S0034-4257(00)00098-5.

[13] N. Desmars, F. Bonnefoy, S. T. Grilli, G. Ducrozet, Y. Perignon, C.-A. Guérin, P. Ferrant, Experimental and numerical assessment of deterministic nonlinear ocean waves prediction algorithms using non-uniformly sam-

pled wave gauges, Ocean Engineering 212 (2020) 107659. doi:10.1016/j.oceaneng.2020.107659.

[14] G. Evensen, The ensemble Kalman filter: Theoretical formulation and practical implementation, Ocean Dynamics 53 (2003) 343–367.

[15] B. Kerbl, G. Kopanas, T. Leimkühler, G. Drettakis, 3d Gaussian splatting for real-time radiance field rendering, in: ACM SIGGRAPH, 2023, pp. 1–14.

[16] M. A. Donelan, W. M. Drennan, A. K. Magnusson, Nonstationary analysis of the directional properties of propagating waves, Journal of Physical Oceanography 26 (1996) 1901–1914.

[17] F. Fucile, G. Bulian, C. Lugni, A probabilistic approach for the quantification of prediction error in deterministic phase-resolved wave forecasting, Ocean Engineering 163 (2018) 718–736. doi:10.1016/j.oceaneng.2018.04.079.

[18] P. Naaijen, K. Trulsen, E. Blondel-Couprie, Limits to the extent of the spatio-temporal domain for deterministic wave prediction, International Shipbuilding Progress 61 (2014) 203–223. doi:10.3233/ISP-140113.

[19] U. D. Nielsen, Estimations of on-site directional wave spectra from measured ship responses, Marine Structures 19 (2006) 33–69.

[20] B. R. Hunt, E. J. Kostelich, I. Szunyogh, Efficient data assimilation for spatiotemporal chaos: a local ensemble transform Kalman filter, Physica D: Nonlinear Phenomena 230 (2007) 112–126. doi:10.1016/j.physd.2006.11.008.

[21] H. L. Tolman, User manual and system documentation of WAVEWATCH III version 3.14, Technical Report 276, NOAA/NWS/NCEP/MMAB, 2009.

[22] F. Wang, Q. Zhu, C. Cai, X. Wang, R. Qiao, From points to waves: Fast ocean wave spatial–temporal fields estimation using ensemble transform Kalman filter with optical measurement, Coastal Engineering 197 (2025) 104690. doi:10.1016/j.coastaleng.2024.104690.

[23] I. R. Young, On the measurement of directional wave spectra, Applied Ocean Research 16 (1994) 283–294.

[24] J. Capon, High-resolution frequency-wavenumber spectrum analysis, Proceedings of the IEEE 57 (1969) 1408–1418.

[25] R. O. Schmidt, Multiple emitter location and signal parameter estimation, IEEE Transactions on Antennas and Propagation 34 (1986) 276–280.

[26] C. E. Rasmussen, C. K. I. Williams, Gaussian Processes for Machine Learning, MIT Press, 2006.

[27] M. K. Titsias, Variational learning of inducing variables in sparse Gaussian processes, in: Proceedings of the 12th International Conference on Artificial Intelligence and Statistics (AISTATS), 2009, pp. 567–574.

[28] M. Liu, S. Dong, J. Chen, Y. Zhang, Optimized quiescent period prediction under harsh sea states using a linear wave model based on physics-informed neural networks, Ocean Engineering 310 (2024) 118763. doi:10.1016/j.oceaneng.2024.118763.

[29] F. Mohaghegh, J. Murthy, M.-R. Alam, Rapid phase-resolved prediction of nonlinear dispersive waves using machine learning, Applied Ocean Research 117 (2021) 102920. doi:10.1016/j.apor.2021.102920.

[30] S. Ehlers, M. Klein, A. Heinlein, M. Wedler, N. Desmars, N. Hoffmann, M. Stender, Machine learning for phase-resolved reconstruction of nonlinear ocean wave surface elevations from sparse remote sensing data, Ocean Engineering 288 (2023) 116059. doi:10.1016/j.oceaneng.2023.116059.

[31] E. J. Candès, J. Romberg, T. Tao, Robust uncertainty principles: Exact signal reconstruction from highly incomplete frequency information, IEEE Transactions on Information Theory 52 (2006) 489–509.

[32] S. G. Mallat, Z. Zhang, Matching pursuits with time-frequency dictionaries, IEEE Transactions on Signal Processing 41 (1993) 3397–3415.

[33] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, in: Proceedings of the 3rd International Conference on Learning Representations (ICLR), 2015.

[34] K. Hasselmann, T. P. Barnett, E. Bouws, et al., Measurements of wind-wave growth and swell decay during the Joint North Sea Wave Project (JONSWAP), Ergänzungsheft zur Deutschen Hydrographischen Zeitschrift A8 (1973) 1–95.

[35] M. Born, E. Wolf, Principles of Optics: Electromagnetic Theory of Propagation, Interference and Diffraction of Light, 7th ed., Cambridge University Press, 1999.

[36] G. M. Hale, M. R. Querry, Optical constants of water in the 200-nm to 200-μm wavelength region, Applied Optics 12 (1973) 555–563.

[37] Coastal Data Information Program, CDIP wave observation archive, scripps institution of oceanography, university of california san diego, THREDDS/OPeNDAP archive, `https://thredds.cdip.ucsd.edu/thredds/catalog.html`, 2026. Directional moment variables `waveA1Value`, `waveB1Value` and `waveEnergyDensity`; 24 stations, 25,426 quality-flagged records accessed 2026-08-03.